Tenure: software requirements specification

The first version: what Tenure must do, and how well. The structure follows ISO/IEC/IEEE 29148, kept lean.

Document	Software requirements specification
Version	0.1, a working draft
Date	7 November 2026
Status	Approved scope (`docs/milestone-2/FIRST-VERSION.md`); being built
Sources	`docs/milestone-2/answers.md` (Uzair's answers, which win), `docs/ARCHITECTURE.md` (the contract), `packages/copy/COPY.md` (every word)

Revision history

Version	Date	What changed
0.1	7 Nov 2026	First draft: the scope, requirements, use cases, data, architecture, build plan, risks, acceptance tests, traceability and open issues for the first version

Every requirement has a stable ID. Code, tests and commits point back to it: each test's name starts with the ID it covers, for example `FR-KIOSK-03 a second check-in within the hour is not stored`.

Priorities use MoSCoW:

- **Must:** in the first version.
- **Should:** in the first version if time allows.
- **Could:** the polish that follows.
- **Won't:** not in this version.

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1. Introduction

1.1 Purpose

This document says what the first version of Tenure must do (§3) and how well (§4). It is the working document for the people and agents who build, test and review that version, and for Uzair, who approves its scope.

Tenure helps independent gyms keep new members past their first 90 days. It counts each new member's visits week by week, shows who is slipping, and lets a coach nudge them with a message sent from the coach's own phone.

1.2 Scope

The first version is the whole MVP, built simply: every core screen, the account basics, and a simple public website with sign-up. Its core flows are tested end to end, through the server and in the browser, and small rough edges are allowed. It runs on one machine, locally or in Docker. In stall mode it serves phones and the kiosk on that machine's local network (FR-OPS-01).

In the first version

Area	What's in	Where
Accounts	Sign-up with the gym's country, log-in, 30-day sessions, password reset, and staff invites by email	§3.1, FR-STAFF-03
The gym and its staff	Settings, owners and coaches, deactivating and reactivating staff	§3.2, §3.3
Members	The Members list and the member page: add, edit, archive, restore and delete	§3.4, §3.5
The first 90 days	Today and the board	§3.6, §3.7
Nudges	The composer and its Contact menu, the nudges log, and templates	§3.8, §3.9
The kiosk	Check-in by phone or name, joining, and the welcome	§3.10
The import	Members or check-ins from a CSV file	§3.11
Reports and Billing	Cohorts, visits by week, nudge results, and the pilot day	§3.12, §3.13
The shell	Me, the navigation and the two themes	§3.14, §3.15
The website	Home, Pricing, sign-up, and "Try the demo" at the stall	§3.16
Across the product	Five markets, live updates, privacy and deletion	§3.17 to §3.19
Running it	Stall mode, Docker, backup and restore, the demo reset, migrations, and the development outbox	§3.20, §3.21
Quality	Speed, accessibility, security, privacy, reliability, browsers, maintenance, markets and copy	§4

Not in the first version

What	Where it stands
Real email, texts and payments	Stand-ins only, and email lands in the outbox (FR-DEV-02). Real providers come with the hardening, once Uzair has made the accounts (ARCHITECTURE §13)

What	Where it stands
Online hosting	Local and Docker only (FR-OPS-01). Hosting also comes with the hardening
Email verification	Later (FR-AUTH-10). Password reset is in (FR-AUTH-07)
Translations	English only (FR-GLOBAL-06). Translations are decided per market (ARCHITECTURE §2)
Native apps	Tenure runs in Chrome and Safari (NFR-COMP-01)
Links to other gym software	Members and check-ins come in through the CSV import (FR-IMPORT-01) and the kiosk
The add-to-home-screen kiosk	Later (FR-KIOSK-09). The kiosk runs in the browser
A price	The pilot is free, and no price is shown until Uzair sets one (FR-BILL-01, FR-WEB-02)
Analytics and tracking	None (FR-PRIV-01)

The rare edge cases and the rest of the old app's tests wait for the polish that follows (ARCHITECTURE §13).

1.3 Definitions

Term	Meaning
Gym	One customer of Tenure, with its own staff, members, country and time zone. Every read and write is held to the signed-in user's gym (FR-PRIV-02).
Staff	The people who sign in to a gym: its owners and coaches (FR-STAFF-01).
Owner	A staff member with the owner role. Whoever signs the gym up is its first owner (FR-AUTH-03). Owners see every member and have the owner-only pages (FR-STAFF-06).
Co-owner	Another owner, invited with the Owner choice in "Invite staff" (FR-STAFF-03, FR-STAFF-04). A co-owner has the same rights as the first owner. The last active owner can't be deactivated (FR-STAFF-05).
Coach	A staff member with the coach role. A coach sees and acts on only their own members (FR-STAFF-07).
Own members	The members whose coach is that staff member. An owner can coach members too.
Member	Someone who trains at the gym. Members don't sign in: they check in at the kiosk, and staff add or import them (FR-MEM-05, FR-IMPORT-01). A member has a first name, an optional last name (FR-GLOBAL-04) and a joined date, and may have a phone and a coach.
Kiosk	The check-in screen at `/kiosk/:token`, open on the front-desk tablet or screen. It needs no sign-in (FR-KIOSK-01). Members check in there by phone or name, and visitors join (FR-KIOSK-02, FR-KIOSK-04).
Kiosk token	The gym's secret in the kiosk link: anyone with the link can use the kiosk. A bad token shows a not-found screen (FR-KIOSK-01). Settings shows the link and its QR code (FR-GYM-03).
Check-in	One recorded arrival, made at the kiosk or brought in by the import. A second one by the same member within an hour isn't stored (FR-KIOSK-03, FR-IMPORT-07).
Visit	A day on which a member checked in, in the gym's time zone. Weeks, pace and statuses count visits, so several check-ins on one day count once.
The 90-day window	A member's first 90 days: day 0 is the joined date and day 89 the last, in the gym's time zone. The board and Members' "First 90 days" view list only members inside it (FR-BOARD-01, FR-MEM-01). Visits outside it don't count toward weeks, pace or status.
Week	Seven days counted from the joined date. Week 1 is days 0 to 6, and so on to week 13, days 84 to 89, which has only 6. A week is finished once all its days are before today; the current week is still in progress.

Term	Meaning
Status	One of four, worked out from a member's finished weeks. The rules are checked in the order below, and the first that matches wins (ARCHITECTURE §7).
Slipping	Checked first: 2 or more finished weeks, and the last two each under 2 visits.
New	Checked second: fewer than 2 finished weeks.
On the line	Checked third: the last finished week had 3 or more visits.
Below the line	Checked last: everything else.
The line	3 visits a week. Reports draws it on its chart (FR-REPORT-02).
Pace	A member's mean visits a week over their last 3 finished weeks, or over all of them if fewer have finished, to one decimal. There is none before the first finished week.
Nudge	A staff member's message to a member, to bring them back. It is written in the composer, from a template or from scratch (FR-NUDGE-01), and recorded when the sender picks how to send it (FR-NUDGE-02). Tenure never sends it (FR-NUDGE-05).
Channel	How a nudge went out, stored with it: `text`, `whatsapp`, `copy` or `in-person`, shown as "Text", "WhatsApp", "Copied" or "In person" (FR-NUDGE-06). A nudge from before the first version shows none.
Came back	A nudge came back when the member checks in strictly after it and within 7 days (FR-NUDGE-07).
Template	A named nudge message that fills `{first}`, `{coach}`, `{gym}` and `{days}` (FR-TPL-02). Owners edit templates and coaches read them (FR-TPL-01). A new gym gets the three default ones (FR-TPL-03).
Pilot	A gym's free first 90 days on Tenure, counted from the day it signed up. Billing shows "Pilot day {n} of 90" (FR-BILL-01). There is no price.
Archive	Hiding a member and keeping their history. An archived member is left out of every list except Members' "Archived" view, and out of the kiosk's name search (FR-MEM-01, FR-KIOSK-02). Their history still counts in Reports (FR-REPORT-01). At the kiosk, a typed phone or a join that matches them shows "Please see the front desk to check in." (FR-KIOSK-06). "Restore member" or a matching import row brings them back (FR-MEMBER-06, FR-IMPORT-09). Only an archived member can be deleted for good (FR-MEMBER-07).
Stall mode	Tenure as it runs at the expo stall: `npm run stall`, or Docker through `scripts/stall.sh`, serves the built app from the API on the laptop's network (FR-OPS-01). The kiosk, phone and invite links use the laptop's LAN address (FR-GYM-03, FR-STAFF-03), "Try the demo" works (FR-WEB-04), and a button re-seeds the demo gym (FR-OPS-03).
The demo gym	Northside Strength, the seeded sample gym: 3 staff and 85 members, 60 of them in their first 90 days. Its dates are relative to the day it was seeded, and its country follows the laptop's time zone (FR-GLOBAL-05). Its kiosk shows the "Sample data" tag (FR-KIOSK-07). `npm run demo:reset` re-seeds it (FR-OPS-03).
Stand-in	An adapter that keeps an email, a text or a payment on the machine instead of calling a provider. Every environment uses the stand-ins in the first version (FR-DEV-02).
Outbox	The table where the email stand-in puts every email, such as invites and reset links. `/dev/outbox` shows it, in development only (FR-DEV-01).
E.164	The international phone format: "+", the country code, then the number, with no spaces, such as +13125550142. Every phone is stored this way (FR-GLOBAL-02).
MoSCoW	The priority scale in §3: Must, Should, Could and Won't, as defined at the top of this document.

1.4 References

Reference	What it holds
`docs/ARCHITECTURE.md`	The contract: the decisions, layout, dependency rules, design tokens, commands and milestones
`docs/milestone-2/FIRST-VERSION.md`	The approved plan for the first version and for this spec (7 Nov 2026)
`docs/milestone-2/answers.md`	Uzair's product answers, which win over every other source
`docs/milestone-2/kiosk-blend.md`	The kiosk's N1 "Soft corners" blend, with its CSS and contrast checks
`packages/copy/COPY.md`	Every word the product shows
`docs/DEMO.md`	The stall runbook, written in phase 4 (FR-OPS-05)
`legacy/app/SPEC.md`	The old app's product spec and API, the reference for the port
`legacy/app/DEMO.md`	The old app's stall runbook and demo script
`marketing/brand/DESIGN.md`	The brand's voice, for the home page copy round
WCAG 2.1	W3C, Web Content Accessibility Guidelines 2.1, held to level AA (NFR-USE-01)
ISO/IEC/IEEE 29148:2018	Systems and software engineering: life cycle processes, requirements engineering. The shape of this spec
ITU-T E.164	The international public telecommunication numbering plan: the phone format

2. Overall description

2.1 Product perspective

Tenure is a self-contained web app on one machine. In stall mode and in Docker, one server, the API on Node 26, serves the website, the app and the kiosk, answers their JSON requests, and streams live events to open screens (FR-LIVE-01). All data lives in one SQLite file on the same machine (FR-PRIV-01). At the stall that machine is a Mac, running Tenure in Docker or with `npm run stall`, and phones and the kiosk tablet reach it over the laptop's local network (FR-OPS-01).

The browser app is one React app: the website, the app under `/app`, the kiosk, and the invite and reset pages. The website loads without the app's code (FR-WEB-05).

Tenure contacts no outside service (NFR-PRIV-01):

- A nudge opens the coach's own Messages or WhatsApp with the message filled in, and the coach sends it (FR-NUDGE-02, FR-NUDGE-05).
- Email, texts and payments go through stand-in adapters (FR-DEV-02). Email goes to the outbox table. Nothing in the first version sends a text or takes a payment (FR-NUDGE-05, FR-BILL-01).

The first version replaces the old expo app, kept in `legacy/app` as the reference for the port. Its flows and rules carry over into the new layout (ARCHITECTURE §3), changed where answers.md and the first-version plan say so.

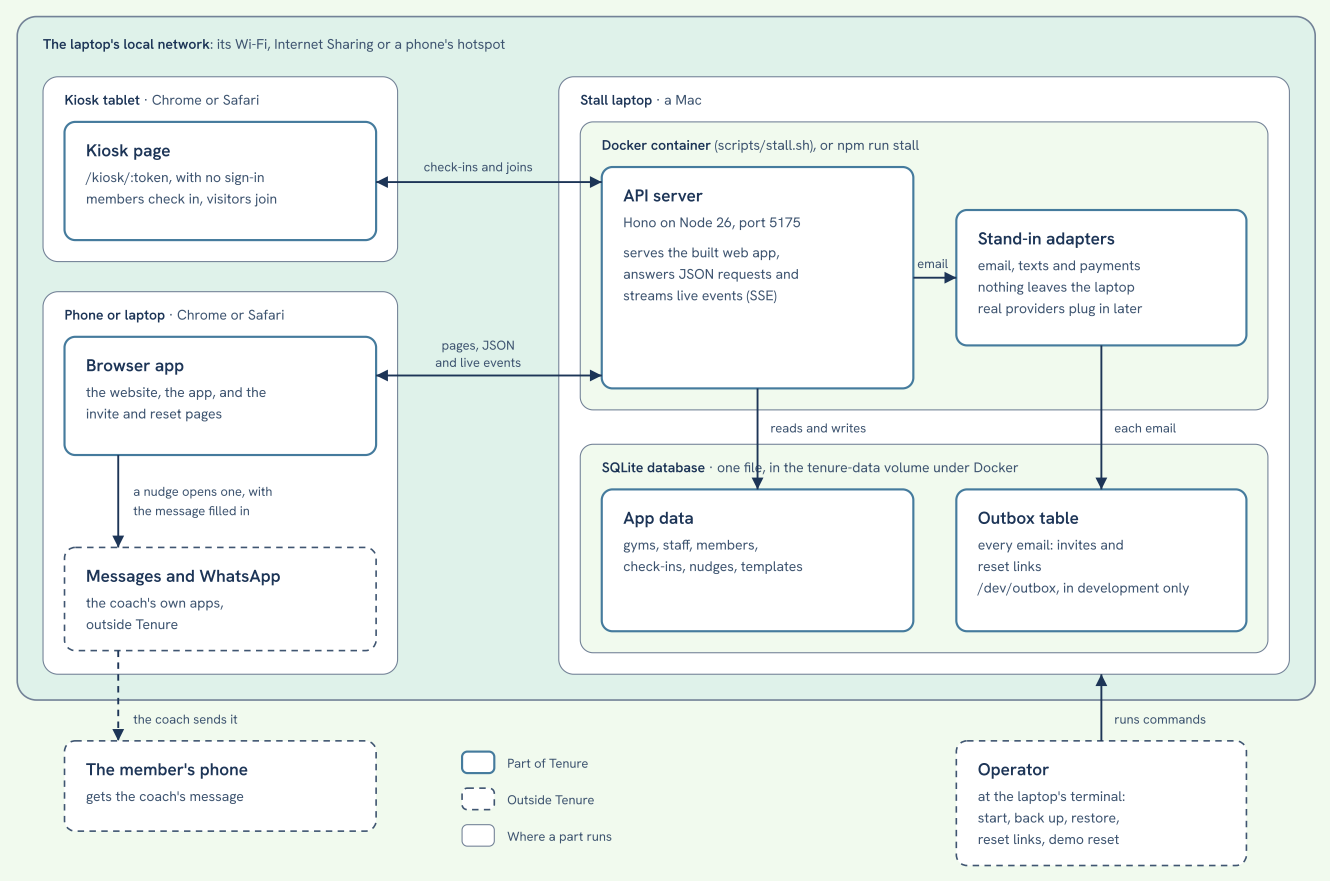


Figure 1. Tenure at the stall. Dashed boxes are outside Tenure. In development (`npm run dev`) the parts run on one Mac without Docker: the web app has its own server on port 5174, and the outbox page is on (FR-DEV-01).

Interfaces

Interface	Between	What passes
HTTP	Browsers on the network, and the API	Pages and JSON. A write must be JSON from a matching Origin (NFR-SEC-02), and bodies are capped (NFR-SEC-05)
Live events	The API and each signed-in tab	One server-sent event stream per tab, scoped to the user (FR-LIVE-02)
Message links	The browser app and the coach's own apps	The member's number and the filled-in message, through text and WhatsApp links (FR-NUDGE-02, FR-NUDGE-03)
The clipboard	The browser app and the device	A copied message or link (FR-NUDGE-02, FR-GYM-03, FR-STAFF-03)
QR codes	Settings and a phone's camera	The kiosk and phone links (FR-GYM-03)
CSV files	The owner's files and the import	Members or check-ins, up to 2 MB (FR-IMPORT-01)
Adapter ports	The API and its providers	Emails to the outbox now; Resend, Twilio and Stripe later (FR-DEV-02)
The terminal	The operator and the machine	The stall, backup, restore, reset-link and demo reset commands (FR-OPS-01 to FR-OPS-03, FR-AUTH-08)

2.2 Product functions

Function	What it does	Requirements
Start a gym	An owner signs up with the gym's country and starts the free pilot, on an empty Today that links to the import and the kiosk	FR-AUTH-01 to FR-AUTH-03, FR-TODAY-03
Check in	Members check in at the kiosk by phone or name; a visitor joins and is checked in at once	FR-KIOSK-02 to FR-KIOSK-06
See who is slipping	Today and the board show each member's weeks, pace and status, and the member page their 90 days, all updated live	FR-TODAY-01, FR-BOARD-01, FR-MEMBER-01, FR-LIVE-01
Nudge	A coach picks a template or writes a message, then contacts the member by text, WhatsApp or a copied message. Tenure records the nudge, and whether the member came back	FR-NUDGE-01 to FR-NUDGE-07, FR-TPL-01
Look after members	Add, edit, archive, restore and delete members, and import members or check-ins from a CSV file	FR-MEM-05, FR-MEMBER-02, FR-MEMBER-05 to FR-MEMBER-07, FR-IMPORT-01
Run the team	Invite coaches and co-owners, deactivate and reactivate them, and keep each coach to their own members	FR-STAFF-02 to FR-STAFF-07
Read the results	Cohorts by join month, visits by week and nudge results; the pilot day on Billing	FR-REPORT-01, FR-BILL-01
Set up	The gym's name, country and time zone, and the kiosk and phone links with their QR codes	FR-GYM-01 to FR-GYM-03
Keep accounts safe	Sessions, password reset and rate limits	FR-AUTH-04 to FR-AUTH-07, NFR-SEC-01, NFR-SEC-03
Work in five markets	Phones, dates and times, and single names, by the gym's country	FR-GLOBAL-01 to FR-GLOBAL-04

Function	What it does	Requirements
Introduce Tenure	The website's Home, Pricing and sign-up, and "Try the demo" at the stall	FR-WEB-01 to FR-WEB-04
Run the stall	Stall mode and Docker, backup and restore, the demo reset, and reset links	FR-OPS-01 to FR-OPS-03, FR-AUTH-08

2.3 User classes

User class	Who	What they use	What they may do
Owner	Runs the gym, and signed it up	The whole app, on a laptop or a phone	Everything in their gym, including the owner-only pages (FR-STAFF-06). Sees every member and every live event (FR-LIVE-02)
Co-owner	A second owner, invited by an owner	The same as the owner	The same as the owner (FR-STAFF-01, FR-STAFF-04)
Coach	Coaches members, and joined through a coach invite	Today, the board, Members, Nudges and Templates, often on their own phone	Works with only their own members (FR-STAFF-07): adds, archives, restores and nudges them (FR-MEM-05, FR-MEMBER-05, FR-MEMBER-06). Reads templates (FR-TPL-01). Gets 403 from owner routes (FR-STAFF-06)
Member at the kiosk	A member, or a newcomer joining	The kiosk only, with no account	Checks in by phone or name, or joins with a first name (FR-KIOSK-02, FR-KIOSK-04). Sees their welcome (FR-KIOSK-05), and the "First L." names the name search shows
Stall visitor	Someone at the expo stall	The kiosk, and "Try the demo" in stall mode	Joins at the kiosk, and uses the demo gym as its owner, with everything usable (FR-WEB-04). The demo is re-seeded between visitors (FR-OPS-03)
Operator	Whoever runs the machine, such as the stall team	The laptop's terminal and the runbook (FR-OPS-05)	Starts Tenure (FR-OPS-01), backs up and restores (FR-OPS-02), re-seeds the demo (FR-OPS-03) and prints reset links (FR-AUTH-08). Only someone at the laptop can get a reset link

2.4 Operating environment

Part	Environment
Browsers	Chrome on laptops and Android; Safari on Mac, iPhone and iPad (NFR-COMP-01). The blended background needs a 2024 or later browser, and older ones show the flat page ground (ARCHITECTURE §2)
Screens	From phone width up: below 1024 px a header and a bottom tab bar, and from 1024 px a sidebar (FR-SHELL-01, FR-SHELL-02)
The server	A Mac with Docker Desktop, running `scripts/stall.sh`, or Node 26 directly, with `npm run stall` (FR-OPS-01). One process serves everything on port 5175, which must be free. The data is one SQLite file, kept in the `tenure-data` volume under Docker
The network	The laptop's local network: its Wi-Fi, Internet Sharing, or a phone's hotspot. Phones and the kiosk join it and open the laptop's LAN address over plain HTTP, so the session cookie isn't marked Secure (NFR-SEC-01). Tenure needs no internet while it runs: it calls no outside service (NFR-PRIV-01), and its fonts are self-hosted. The first Docker build does, to fetch packages (ARCHITECTURE §12)
The kiosk	A tablet at the front desk, such as an iPad in Safari, or a second browser window. It works by touch or with a hardware keyboard (FR-KIOSK-08)

Part	Environment
Rate limits under Docker	Every device counts as one for the rate limits (NFR-SEC-03), because Docker Desktop's port publishing hides each client's address (ARCHITECTURE §12)
Development	`npm run dev` on a Mac with Node 26: the API on port 5175 and the web app on 5174, with hot reload and the outbox page (FR-DEV-01)

2.5 Constraints

Constraint	What it means	Source
The contract	The layout, dependency rules, design tokens and commands in ARCHITECTURE.md hold. To break a rule, that file changes first, in its own commit. The lint checks the dependency rules (NFR-MAINT-01)	ARCHITECTURE §3, §4, §9
The stack	A TypeScript monorepo: React 19 and Vite, Hono on Node 26, SQLite through Drizzle, npm workspaces and Turborepo. Node 26 runs the TypeScript directly, and nothing is installed globally	ARCHITECTURE §2, CLAUDE.md
One test-only package	`@playwright/test` is the only package added for the first version (NFR-MAINT-04)	FIRST-VERSION.md
The gates	Typecheck, lint, test, build and e2e pass before each phase is done (NFR-MAINT-02). Every Must has a test that names its ID, or a stated check (NFR-MAINT-03)	ARCHITECTURE §11
The copy rules	Every word comes from COPY.md, in sentence case. A line describes the action rather than quoting a button, and button labels take no closing punctuation (NFR-COPY-01). COPY.md's own rules hold too: numerals always, and no exclamation marks	ARCHITECTURE §10, COPY.md
Five markets	Only the US, Canada, the UK, Pakistan and the UAE can be chosen (FR-GLOBAL-01), and the interface is in English only (FR-GLOBAL-06)	ARCHITECTURE §2, answers.md
Nothing leaves the machine	No outside service is contacted (NFR-PRIV-01), and Tenure sends no message itself (FR-NUDGE-05)	ARCHITECTURE §2
Printed links	`/kiosk/:token` and `/invite/:token` keep their shape, so printed links still work	ARCHITECTURE §8
Hands off	`legacy/app` and `marketing/` don't change during the port	CLAUDE.md

2.6 Assumptions and dependencies

Assumptions

- The laptop's clock and time zone are right on the day. The demo gym's dates are relative to the day it was seeded, in the laptop's zone, so it is re-seeded on the morning of the stall (FR-GLOBAL-05, ARCHITECTURE §12).
- The operator re-seeds the demo between visitors (FR-OPS-03). Until then, a visitor who taps "Try the demo" sees what earlier visitors entered at the kiosk, names and phones included, as any owner would (FR-WEB-04).
- The laptop's firewall lets phones and the kiosk connect on port 5175.
- Coaches send nudges from their own phone or computer, with Messages or WhatsApp set up on it. A text needs mobile service, and WhatsApp needs the internet (FR-NUDGE-05).
- Each gym is in one of the five markets (FR-GLOBAL-01). Its members' phones are local numbers, or typed with "+" (FR-GLOBAL-02). The demo keeps its +1 sample numbers in every country

(FR-GLOBAL-05).

- A gym's existing members come in as a CSV file of up to 2 MB, or as pasted text (FR-IMPORT-01).
- No email leaves the machine. Staff take invite links by copying them (FR-STAFF-03), and reset links from the operator (FR-AUTH-08).
- Backups are the operator's to keep. A deleted member stays in older backups (NFR-PRIV-02).

Dependencies

Dependency	What relies on it
Node 26 and npm, with internet for the first install	Every command (ARCHITECTURE §12)
Docker Desktop on the Mac	`scripts/stall.sh`, and the `tenure-data` volume that backups copy (FR-OPS-01, FR-OPS-02)
`libphonenumber-js` and its metadata	Which numbers are real in each country, and their E.164 form (FR-GLOBAL-02)
The Intl data in Node and the browsers	Dates and times in the gym's locale, with en-001 where a country has no English formats (FR-GLOBAL-03)
The device's own apps	Text and WhatsApp links open Messages and WhatsApp with the message filled in. iOS opens WhatsApp only on the tap itself, so each menu item is a link (FR-NUDGE-02, FR-NUDGE-03)
Playwright's browsers	The e2e gate: Chrome through the installed browser, and WebKit downloaded once. If WebKit has no build for this macOS, iPhone Safari on the Simulator is checked by hand at each pause (NFR-COMP-01)
Uzair's picks	The design polish picks before the screens are built, and the home page words before the website (FR-WEB-01)
The core-flow edge-case answers	They fold into the backend jobs before those merge

3. Functional requirements

Words in quotes are the interface's own words, taken from COPY.md or picked by Uzair.

3.1 Accounts and sign-in (AUTH)

ID	Requirement	Priority
FR-AUTH-01	An owner shall create a gym at `/signup` with the gym's name, their own name, email, a password of 8 to 200 characters and the gym's country. The browser's time zone is sent with the form.	Must
FR-AUTH-02	The Country choice shall list the five open markets (FR-GLOBAL-01). It is preset when the detected country is one of them and otherwise left empty. A missing or unlisted country is refused: "Choose a country from the list."	Must
FR-AUTH-03	Sign-up shall create the gym, its owner and the three default templates, then sign the owner in to an empty Today. The line under the title reads "Start a free 90-day pilot for your gym."	Must
FR-AUTH-04	Staff shall log in with email and password. A wrong email or password gets one message, and an unknown email takes as long to answer as a wrong password.	Must
FR-AUTH-05	A session shall last 30 days, in an httpOnly cookie. Log-out ends it, and an expired session is refused.	Must
FR-AUTH-06	A deactivated user shall be refused at log-in, and their sessions and live stream end at once.	Must
FR-AUTH-07	Password reset, from "Forgot your password?" on Login: ▪ `/reset` asks for the email ("Reset your password", button "Send reset link") and gives the same answer whether or not the email exists; ▪ the emailed link opens a second page ("Choose a new password", button "Save password"); ▪ the link works once and expires after an hour; ▪ saving ends the user's other sessions.	Must
FR-AUTH-08	On the stall laptop and in Docker, `npm run reset-link -- <email>` shall print a reset link, since the outbox page is development only (FR-DEV-01).	Must
FR-AUTH-09	A signed-in visitor to `/login` or `/signup` shall go straight to `/app`.	Should
FR-AUTH-10	Email verification.	Won't

3.2 Gym settings (GYM)

ID	Requirement	Priority
FR-GYM-01	Owners shall edit the gym's name, country (one of the five) and time zone at `/app/settings`. Saving updates the signed-in gym everywhere.	Must
FR-GYM-02	The time-zone list shall give US gyms the old app's 7 zones (plus the current one if it is missing), and every other country all of its zones by city, east to west.	Must
FR-GYM-03	Settings shall show the kiosk link and the phone link, each with a QR code and a Copy button. In stall mode they use the laptop's LAN address.	Must
FR-GYM-04	Settings shall link to the import page with "Import members or check-ins".	Must
FR-GYM-05	A settings change shall publish an event, so open screens show it.	Should

3.3 Staff and roles (STAFF)

ID	Requirement	Priority
FR-STAFF-01	The roles shall be owner and coach. A gym may have several owners.	Must
FR-STAFF-02	Owners shall see the staff list: each person's role, their members in the 90-day window, how many are slipping (in the attention colour), and an "Inactive" tag where it applies.	Must
FR-STAFF-03	Owners shall invite staff through "Invite staff", choosing Coach or Owner. The invite is emailed (to the outbox) and its link is also shown to copy. It works once, for 7 days, and uses the LAN link in stall mode.	Must
FR-STAFF-04	`/invite/:token` shall let the invitee set their name and password and join the gym, with the owner wording on an owner invite. A bad or used link shows "This link doesn't work". Accepting ends any old session in that browser.	Must
FR-STAFF-05	Owners shall deactivate staff after a confirm, and bring them back with "Reactivate". Their history stays. The last active owner can't be deactivated.	Must
FR-STAFF-06	These shall be owner only: Reports, Settings, the import, Billing, member edits (name, coach, phone), the template editor, the coach filter and column, and the pilot day. A coach gets 403 from their routes.	Must
FR-STAFF-07	A coach shall see and act on only their own members. Another coach's member answers 404.	Must

3.4 Members (MEM)

ID	Requirement	Priority
FR-MEM-01	`/app/members` shall offer one three-way choice: "First 90 days" (the default), "Everyone" (every member who isn't archived) and "Archived".	Must
FR-MEM-02	Members and the board shall share a filter bar, kept in the URL: ▪ search, folding case and accents, with digit queries ignoring leading zeros; ▪ status chips, each with its status dot; ▪ for owners, a coach filter whose options come from the members on screen.	Must
FR-MEM-03	The list shall be a table from 48 rem and cards below. Slipping names keep the attention colour. Names are plain text, and each row ends with a small quiet "Open" button whose accessible name includes the member ("Open Sam O.").	Must
FR-MEM-04	Names shall show as "First L.", or as a single name alone ("Leo"), sorted with the gym's locale; members with no last name come first.	Must
FR-MEM-05	Owners and coaches shall add a member by hand: "Add member" opens "Add a member", with a first name, an optional last name, phone, joined date (today or earlier) and coach. A coach's new member is theirs, and a taken phone is refused.	Must
FR-MEM-06	The list shall have an empty state with filters set, and one without: "Members are listed here once they join at the front desk or are imported."	Must

3.5 The member page (MEMBER)

ID	Requirement	Priority
FR-MEMBER-01	`/app/members/:id` shall show the member's facts (name, coach, phone, joined date), a days grid of the 90 days, their check-ins and their nudges with channel words. An unknown member shows a not-found view.	Must

ID	Requirement	Priority
FR-MEMBER-02	Owners shall change a member's name (first name required, last name optional), coach ("No coach" is an option) and phone (read in the gym's country). Each change sends only its own field, and shows the server's error under it.	Must
FR-MEMBER-03	The phone shall show in national format when its calling code matches the gym's country, and international otherwise.	Must
FR-MEMBER-04	"Nudge →" in the page's head row shall open the composer (FR-NUDGE-01).	Must
FR-MEMBER-05	"Archive", confirmed by "Archive this member?" and its "Archive member" button. Owners archive any member and coaches their own.	Must
FR-MEMBER-06	An archived member's page shall offer "Restore member", with the same permissions.	Must
FR-MEMBER-07	An owner shall permanently delete an archived member: "Delete permanently", confirmed by "Delete this member for good?" (which says their visits and nudges go too) and its "Delete member" button. Their check-ins and nudges are deleted with them.	Must

3.6 Today (TODAY)

ID	Requirement	Priority
FR-TODAY-01	Today (`/app`) shall show: ▪ four KPIs: new members, the share on the line, slipping, and nudged this week; ▪ "Slipping now"; ▪ "Checked in today", with times in the gym's locale; ▪ a short board. Coaches see only their own members.	Must
FR-TODAY-02	"Slipping now" shall list the viewer's own members first, then by most days since the last visit, each with "Nudge →". A member with no coach shows the last-visit words alone.	Must
FR-TODAY-03	An owner's empty gym shall show "Import members" (to the import page) and "Open the kiosk".	Must
FR-TODAY-04	Today shall update live: a check-in appears, with a toast.	Must
FR-TODAY-05	Today shall link to the board.	Should

3.7 The board (BOARD)

ID	Requirement	Priority
FR-BOARD-01	`/app/board` shall list every member in their first 90 days, slipping first and then the newest, with the filter bar (FR-MEM-02), 13 week marks, pace, status and the legend.	Must
FR-BOARD-02	Every board row shall end with "Open". The first slipping row of Today's board also gets "Nudge →".	Must
FR-BOARD-03	A new check-in shall fill its week cell with a pop, scaling from 1.25 to 1 over 320 ms. Nothing moves under reduced motion.	Must

3.8 Nudges (NUDGE)

ID	Requirement	Priority
FR-NUDGE-01	The composer ("Nudge {first}") shall offer a template, or "Write my own", and an editable message.	Must

ID	Requirement	Priority
FR-NUDGE-02	One "Contact {first}" button shall open a menu: ▪ "Text message" (on phones only) and "WhatsApp" each open that app with the message filled in; ▪ "Copy message" copies the message. Picking any of them records the nudge with that channel.	Must
FR-NUDGE-03	Each menu item shall be a link that records on the same click, because iOS refuses to open WhatsApp after a delay.	Must
FR-NUDGE-04	A member with no phone shall get "No phone on file" and "Record nudge", recorded as in person.	Must
FR-NUDGE-05	Tenure shall never send a message itself: every message leaves from the coach's own phone or computer.	Must
FR-NUDGE-06	`/app/nudges` shall list the last 30 days of nudges. Each row shows: ▪ the member's plain name and "Open"; ▪ the start of the message; ▪ the date in the gym's locale; ▪ who sent it; ▪ its channel ("Text", "WhatsApp", "Copied" or "In person"; nothing for nudges from before the first version); ▪ the outcome: came back, waiting, or no visit.	Must
FR-NUDGE-07	A nudge "came back" when a check-in comes strictly after it and within 7 days.	Must
FR-NUDGE-08	Recording a nudge shall check that the member and the template belong to the gym (otherwise not found), and publish an event.	Must

3.9 Templates (TPL)

ID	Requirement	Priority
FR-TPL-01	`/app/templates` shall list the templates. Owners create, edit (a name of up to 60 characters, a body of up to 1,000) and delete them through a confirm, with a live preview. Coaches can read them.	Must
FR-TPL-02	Templates shall fill `{first}`, `{coach}` (the sender's first name), `{gym}` and `{days}`.	Must
FR-TPL-03	A new gym shall get the three default templates.	Must

3.10 The kiosk (KIOSK)

ID	Requirement	Priority
FR-KIOSK-01	`/kiosk/:token` shall work without signing in; the token is the gym's secret. A bad token shows a not-found screen.	Must
FR-KIOSK-02	A member shall check in with their phone or by name: ▪ the phone is read in the gym's country, with "+" for others, formatted as it's typed; ▪ the name search matches from the start, folds case and accents, shows up to 6 people as "First L.", and leaves archived members out.	Must
FR-KIOSK-03	A second check-in by the same member within an hour shall be welcomed but not stored.	Must
FR-KIOSK-04	A visitor shall join with a first name (last name and phone optional). A taken phone is refused. They join today, are checked in, and get the next active staff member as coach, in turn.	Must

ID	Requirement	Priority
FR-KIOSK-05	The welcome shall say: ▪ "Welcome, {first}. Day 1 of 90." on a first visit; ▪ "Welcome back, {first}. Week {n}." after that; ▪ "Welcome back, {first}." from day 90. It adds the visits line, and the line enters with a pop.	Must
FR-KIOSK-06	A typed phone, or a join, that matches an archived member shall show "Please see the front desk to check in." and record nothing.	Must
FR-KIOSK-07	The kiosk's look: ▪ the N1 "Soft corners" blend, a different pattern in each theme (`docs/milestone-2/kiosk-blend.md`); ▪ text of 64 px at most, except the typed number at up to 128 px; ▪ 64 px keys and `line-blend` outlines; ▪ the "Sample data" tag on the demo gym.	Must
FR-KIOSK-08	The kiosk shall return to its start after 30 s idle, and work with a hardware keyboard.	Must
FR-KIOSK-09	An add-to-home-screen app.	Won't

3.11 The CSV import (IMPORT)

ID	Requirement	Priority
FR-IMPORT-01	`/app/settings/import` (owner only) shall import members or check-ins, chosen with a segmented choice, from a file of up to 2 MB or pasted text.	Must
FR-IMPORT-02	It shall suggest a column mapping from the file's headers, which the owner can change, and name any required field that's missing: a first name or a full name and a join date for members; a time and an ID or phone for check-ins.	Must
FR-IMPORT-03	A "Full name" column shall be split: the first word becomes the first name and the rest the last name; "Lopez, Maria" reads as Maria Lopez; a single word is a first name only.	Must
FR-IMPORT-04	Dates shall be read in the gym country's order, unless the file's own dates settle it. One line says how they are read, and the first rows show as read. A quiet button flips the order: "Read them day first" or "Read them month first".	Must
FR-IMPORT-05	Problems shall be listed by row ("Row {n}: ..."). A bad phone, email or coach keeps the member and notes the problem. A phone that belongs to another member is left as it was.	Must
FR-IMPORT-06	Members shall match by external ID, then by phone. Unchanged rows are skipped, a blank never overwrites, and duplicates within the file are skipped.	Must
FR-IMPORT-07	Check-ins shall match their member by ID, then by phone. A check-in in the future, before joining, or within an hour of another is skipped.	Must
FR-IMPORT-08	Nothing shall be saved until the owner commits, and the commit writes everything in one transaction. A row's coach must belong to the gym.	Must
FR-IMPORT-09	A row that matches an archived member shall restore them, and the check step says so.	Must
FR-IMPORT-10	The result shall show what was created, updated and skipped, with a toast. Open screens update once, however many rows came in.	Must

3.12 Reports (REPORT)

ID	Requirement	Priority
FR-REPORT-01	`/app/reports` (owner only) shall show cohorts by join month (the share still coming at day 30, 60 and 90), the visits-by-week chart and the nudge figures. Archived members' history still counts.	Must
FR-REPORT-02	The chart: ▪ still coming is solid, in the text colour; ▪ slipped is dashed, in the slipping colour, with point markers; ▪ the 3-a-week line is dotted, in soft text; ▪ each line is labelled at its end, with no legend; ▪ a description is given for screen readers.	Must
FR-REPORT-03	Months and dates shall follow the gym's locale, and nudges shall be counted in the gym's time zone.	Must

3.13 Billing (BILL)

ID	Requirement	Priority
FR-BILL-01	`/app/billing` (owner only) shall show "Pilot day {n} of 90" (1 to 90), the 90 days as one bar, and that the pilot is free. No price and no plan button.	Must

3.14 Me (ME)

ID	Requirement	Priority
FR-ME-01	`/app/me` shall show the user's name, email, gym and role, the theme choice (a segmented choice) and "Log out".	Must

3.15 The app shell (SHELL)

ID	Requirement	Priority
FR-SHELL-01	Below 1024 px: ▪ a header shows the gym name, the theme toggle and "Log out", with no logo; ▪ a bottom tab bar holds Today, Board, Members, Nudges and More; ▪ More opens a sheet with the rest of the role's pages, and stays marked while one of them is open.	Must
FR-SHELL-02	From 1024 px, a sidebar shall hold the logo, the gym name and every page, with the theme toggle and "Log out" at its foot.	Must
FR-SHELL-03	Owners shall see "Pilot day {n} of 90" under the gym name in the sidebar, and after it in the 768 to 1023 px header. Coaches and phones don't.	Must
FR-SHELL-04	Owner pages shall appear only in owners' navigation. A signed-out visitor is sent to `/login`.	Must
FR-SHELL-05	A screen whose data fails to load shall offer "Try again".	Must
FR-SHELL-06	After a navigation, focus shall move to the page's title, shown by a short focus bar.	Must
FR-SHELL-07	The themes shall be Navy & Lime (dark) and Summer Ocean Breeze (light). The theme follows the device, the toggle's choice is saved on it, and the page never flashes the wrong theme.	Must

3.16 The website (WEB)

ID	Requirement	Priority
FR-WEB-01	Home (`/`) shall offer a free pilot, with the words Uzair picks in the home page copy round.	Must
FR-WEB-02	Pricing (`/pricing`) shall describe the free 90-day pilot and what it includes, with no price.	Must
FR-WEB-03	The website shall lead to sign-up (FR-AUTH-01).	Must
FR-WEB-04	In stall mode, "Try the demo" shall sign a visitor in as the demo gym's owner, with everything usable.	Must
FR-WEB-05	The website shall load without the app's code.	Must

3.17 Global-ready (GLOBAL)

ID	Requirement	Priority
FR-GLOBAL-01	Five markets shall be open: the US, Canada, the UK, Pakistan and the UAE.	Must
FR-GLOBAL-02	Phones: ▪ a number without "+" is read in the gym's country; ▪ a "+" number is international, unless its digits also make a real local number, which then wins; ▪ one rule on the server reads every phone, and every number is stored as E.164.	Must
FR-GLOBAL-03	Dates and times shall follow the gym's locale and time zone: English for the gym's country where the browser has it, otherwise international English (en-001), each with the country's 12- or 24-hour clock.	Must
FR-GLOBAL-04	A member may have a single name, everywhere.	Must
FR-GLOBAL-05	The demo gym's country shall come from the laptop's time zone when it's one of the five, otherwise the US. `TENURE_COUNTRY` overrides it, and the demo's +1 sample numbers stay.	Must
FR-GLOBAL-06	The interface shall be in English only.	Must

3.18 Live updates (LIVE)

ID	Requirement	Priority
FR-LIVE-01	Every write shall publish an event, and open screens refresh: a check-in, a join, a member edited, archived, restored or deleted, a nudge, an import, a staff change, a settings change.	Must
FR-LIVE-02	Owners shall receive every event and coaches only their own members'. A member moved to another coach reaches both coaches.	Must
FR-LIVE-03	Refreshes shall be batched: an import of 135 check-ins refreshes Today once, and so does a reconnect.	Must
FR-LIVE-04	The live stream shall end when the session ends or the server stops.	Must

3.19 Privacy and data (PRIV)

ID	Requirement	Priority
FR-PRIV-01	All data shall stay on the machine running Tenure. There is no analytics or tracking.	Must
FR-PRIV-02	Every read and write shall be held to the user's own gym. An id from another gym is not found.	Must
FR-PRIV-03	A member's data shall be deletable on request (FR-MEMBER-07).	Must

3.20 Operations (OPS)

ID	Requirement	Priority
FR-OPS-01	`npm run stall`, and `scripts/stall.sh` through Docker, shall serve everything on the laptop's network, for phones and the kiosk tablet.	Must
FR-OPS-02	`npm run backup` shall copy the database to a dated file. `npm run restore -- <file>` restores one, and refuses while the server runs or when the file fails its integrity check. `scripts/backup.sh` does both for the Docker volume.	Must
FR-OPS-03	`npm run demo:reset`, and a button in stall mode, shall re-seed the demo gym between stall visitors.	Must
FR-OPS-04	Migrations shall run on start, so an older database upgrades.	Must
FR-OPS-05	`docs/DEMO.md` shall be the stall's runbook.	Should

3.21 Development (DEV)

ID	Requirement	Priority
FR-DEV-01	`/dev/outbox` shall show the stand-in emails, in development only (`npm run dev`).	Must
FR-DEV-02	Email, texts and payments shall go through adapters with stand-ins. Real providers (Resend, Twilio, Stripe) plug in later.	Must

4. Non-functional requirements

ID	Requirement	How it's verified
NFR-PERF-01	Screens shall load in under 2 s on the stall laptop with the demo gym's 85 members.	Playwright timing in each screen's flow
NFR-PERF-02	A kiosk check-in shall answer in under 1 s.	Playwright timing in the kiosk flow
NFR-PERF-03	A live update shall reach other open screens in under 2 s.	Playwright, with two browsers: a kiosk check-in, then the owner's Today
NFR-PERF-04	A 2,000-row import's check step shall show in under 5 s.	Playwright, with a generated file
NFR-USE-01	Every colour pair shall meet WCAG 2.1 AA, and every line and focus ring 3:1.	`contrast.test.ts`
NFR-USE-02	The core flows shall work with a keyboard alone.	Keyboard steps in the Playwright flows
NFR-USE-03	Touch targets shall be at least 44 px, and 64 px on the kiosk.	Component tests, plus a Playwright size check
NFR-USE-04	Nothing shall move when the device asks for reduced motion.	Playwright with reduced motion emulated
NFR-USE-05	Colour shall never be the only cue: dots, dashes, markers, words and edges carry meaning too.	Review at each pause
NFR-SEC-01	Passwords shall be stored with scrypt and session tokens as hashes. The cookie is httpOnly and SameSite=Lax, and Secure over HTTPS.	API tests
NFR-SEC-02	A write shall be refused unless it is JSON from a matching Origin.	API tests
NFR-SEC-03	Rate limits: log-in 10 failures per account per 10 minutes; sign-up 5 per 10 minutes; reset requests limited; the kiosk 30 a minute.	API tests
NFR-SEC-04	Every route shall declare whether it is public, signed-in or owner only. One sweep test checks every route against its declaration.	`permissions.test.ts`
NFR-SEC-05	Request bodies shall be capped at 64 KB, and at 4 MB for the import.	API tests
NFR-SEC-06	Nothing secret shall reach the browser. The outbox page is off outside development.	Config review, plus an API test
NFR-PRIV-01	Tenure shall contact no outside service. Nudges open the coach's own apps.	Review, plus a test that the stand-ins make no network call
NFR-PRIV-02	A deleted member shall leave the live database. Older backups still hold them, and DEMO.md says so.	API test, plus the runbook
NFR-REL-01	The server shall stop cleanly within about a second (a 3 s fallback), never held up by an open stream.	`shutdown.test.ts`
NFR-REL-02	Migrations shall upgrade a database from any earlier migration to the latest.	A db test from 0000
NFR-REL-03	A backup taken while the server writes shall restore with the same row counts.	A backup test
NFR-REL-04	The Docker container shall report its health.	The compose health check

ID	Requirement	How it's verified
NFR-COMP-01	Tenure shall work in Chrome (laptops and Android) and Safari (Mac, iPhone and iPad).	Playwright's Chrome and WebKit projects, plus iPhone Safari on the Simulator at each pause
NFR-MAINT-01	The dependency rules (ARCHITECTURE §4) shall hold.	The ESLint boundaries in `npm run lint`
NFR-MAINT-02	Five gates shall pass before each phase is done: typecheck, lint, test, build and e2e.	The runner at each phase's merge
NFR-MAINT-03	Every Must shall have a test that names its ID, or a stated manual check.	The trace test
NFR-MAINT-04	Only one package shall be added: `@playwright/test`, test-only.	Lockfile review
NFR-LOC-01	The five markets, en-001 and the phone rule shall behave as FR-GLOBAL says.	Domain tests per country
NFR-COPY-01	Every word shall come from COPY.md, in sentence case, and a line never quotes a button.	The copy tests

5. Use cases

Ten use cases walk through the first version's core flows. Steps cite the requirements they rest on, and the Requirements row lists them all; Appendix A traces them to their tests. An alternative is numbered after the step it leaves from, so 2a leaves step 2. Words in quotes are the interface's own, from COPY.md.

ID	Use case	Actor
UC-01	Sign up and import members	Owner
UC-02	Check in or join at the kiosk	Member at the kiosk
UC-03	See who's slipping and contact them	Coach
UC-04	Work the board and Members	Owner or coach
UC-05	Manage staff	Owner, and the person invited
UC-06	Read Reports	Owner
UC-07	Archive, restore and delete a member	Owner, or a coach for their own members
UC-08	Edit Settings	Owner
UC-09	Reset a password	Staff member, and the operator
UC-10	Run the stall demo	Stall visitor, and the operator

5.1 Sign up and import members (UC-01)

Use case	An owner starts their gym on Tenure and brings in its members from a CSV file
Actor	Owner
Precondition	Tenure is running (FR-OPS-01). The owner has no account. They have the gym's member list as a CSV file of up to 2 MB, or as text to paste.
Main flow	1. The owner follows the website, from Home or Pricing, to sign-up (FR-WEB-01 to FR-WEB-03). 2. At `/signup` they enter the gym's name, their name, their email and a password of 8 to 200 characters, and choose the gym's country from the five. The country is preset when the detected one is among them. The browser sends its time zone with the form (FR-AUTH-01, FR-AUTH-02). 3. They press "Start the pilot". Tenure creates the gym, its owner and the three default templates, and signs the owner in to an empty Today. The session lasts 30 days (FR-AUTH-03, FR-AUTH-05, FR-TPL-03). 4. Today offers "Import members" and "Open the kiosk". The owner presses "Import members" (FR-TODAY-03). 5. On the import page they choose Members, then choose the CSV file or paste it (FR-IMPORT-01). 6. Tenure suggests a column for each field from the file's headers, and the owner changes any that are wrong. A "Full name" column is split into a first and a last name (FR-IMPORT-02, FR-IMPORT-03). 7. One line says how dates are read: in the country's order, unless the file's own dates settle it. The first rows show as read (FR-IMPORT-04). 8. The check step shows the first rows and lists any problems by row. Nothing is saved yet (FR-IMPORT-05, FR-IMPORT-08). 9. The owner presses "Import members". Tenure writes every row in one transaction, with phones read in the gym's country and stored as E.164 (FR-IMPORT-08, FR-GLOBAL-02). 10. The result shows what was created, updated and skipped, with a toast. Open screens refresh once, and Today and the board show the members in their first 90 days (FR-IMPORT-10, FR-LIVE-03, FR-TODAY-01, FR-BOARD-01).

Use case	An owner starts their gym on Tenure and brings in its members from a CSV file
Alternatives	1a. A signed-in visitor who opens `/signup` goes straight to `/app` (FR-AUTH-09). 2a. No country, or one not on the list: refused with "Choose a country from the list." (FR-AUTH-02). 2b. The email already has an account: refused, and nothing is created (§6.1, `users`). 2c. A sixth sign-up from the device within 10 minutes is refused (NFR-SEC-03). 4a. The owner starts with the kiosk instead (UC-02), or adds members by hand (UC-04). 5a. A file over 2 MB is refused (FR-IMPORT-01). 5b. The owner imports check-ins instead. Each row needs a time, and a member ID or a phone. A check-in matches its member by ID, then by phone. One in the future, before the member joined, or within an hour of another is skipped (FR-IMPORT-02, FR-IMPORT-07). 6a. A required field has no column: the page names it (FR-IMPORT-02). 7a. The dates read the wrong way: the owner presses "Read them day first" or "Read them month first" (FR-IMPORT-04). 8a. A row has a bad phone, email or coach: the member is kept, and the problem noted. A phone that belongs to another member is left as it was (FR-IMPORT-05). 8b. A row matches a member already there, by external ID and then by phone: it updates them. Unchanged rows and duplicates within the file are skipped, and a blank never overwrites (FR-IMPORT-06). 8c. A row matches an archived member: the import restores them, and the check step says so (FR-IMPORT-09). 8d. The owner leaves before importing: nothing is saved (FR-IMPORT-08).
Postcondition	The gym exists, with its country, its time zone, its owner and the three default templates. The imported members are stored, or nothing is if the owner left before importing.
Requirements	FR-WEB-01 to FR-WEB-03, FR-AUTH-01 to FR-AUTH-03, FR-AUTH-05, FR-AUTH-09, FR-TPL-03, FR-TODAY-01, FR-TODAY-03, FR-BOARD-01, FR-IMPORT-01 to FR-IMPORT-10, FR-GLOBAL-02, FR-LIVE-03, FR-OPS-01, NFR-SEC-03

5.2 Check in or join at the kiosk (UC-02)

Use case	A member checks in at the front desk, or a newcomer joins and checks in
Actor	Member at the kiosk
Precondition	The gym's kiosk link is open on the front-desk tablet or screen (FR-GYM-03). It needs no sign-in (FR-KIOSK-01).
Main flow	1. The member types their phone number on the keypad, or on a hardware keyboard. It's read in the gym's country, or as an international number after "+", and formatted as it's typed (FR-KIOSK-02, FR-KIOSK-08, FR-GLOBAL-02). 2. They press "Check in". Tenure finds the member by the number and stores the check-in, in under 1 s (FR-KIOSK-02, NFR-PERF-02). 3. The welcome says "Welcome back, {first}. Week {n}." with the visits line, which enters with a pop (FR-KIOSK-05). 4. Open staff screens show the check-in: Today lists it with a toast, and the board fills its week cell with a pop (FR-TODAY-04, FR-BOARD-03, FR-LIVE-01, FR-LIVE-02). 5. After a short countdown ("Ready for the next member in {n}."), the kiosk returns to its start.

Use case	A member checks in at the front desk, or a newcomer joins and checks in
Alternatives	Before step 1: a bad token in the link shows a not-found screen (FR-KIOSK-01). 1a. By name: the member presses "Use my name" and types the start of their first name. Up to 6 members show as "First L.", with case and accents folded and archived members left out. The member picks their name (FR-KIOSK-02). 1b. A newcomer presses "New here? Join", enters a first name, with a last name and phone optional, and presses "Join and check in". Tenure adds them, joined today, with the next active staff member as their coach in turn, and checks them in. The welcome says "Welcome, {first}. Day 1 of 90." (FR-KIOSK-04, FR-KIOSK-05, FR-LIVE-01). 1c. The join has no first name: "Enter your first name." (FR-KIOSK-04). 1d. The join's phone belongs to another member: refused. If that member is archived, the kiosk shows "Please see the front desk to check in." and records nothing (FR-KIOSK-04, FR-KIOSK-06). 2a. No member has that number: "We couldn't find that number. Try your name, or join as a new member." Nothing is stored (FR-KIOSK-02). 2b. The number belongs to an archived member: "Please see the front desk to check in." Nothing is recorded. Staff restore the member by hand (UC-07), and they check in again (FR-KIOSK-06, FR-MEMBER-06). 2c. The member checked in within the last hour: they're welcomed, and no second check-in is stored (FR-KIOSK-03). 3a. On a first visit the welcome says "Welcome, {first}. Day 1 of 90.", and from day 90 "Welcome back, {first}." (FR-KIOSK-05). Any step: 30 s without input returns the kiosk to its start (FR-KIOSK-08). Any step: more than 30 requests a minute from the device get "Too many tries. Try again in a minute." (NFR-SEC-03).
Postcondition	The member has one more check-in, unless they had checked in within the hour, weren't found, or are archived. A newcomer is a member, joined today, with a coach and their first check-in. Staff screens show it within 2 s (NFR-PERF-03).
Requirements	FR-KIOSK-01 to FR-KIOSK-06, FR-KIOSK-08, FR-GYM-03, FR-GLOBAL-02, FR-TODAY-04, FR-BOARD-03, FR-MEMBER-06, FR-LIVE-01, FR-LIVE-02, NFR-PERF-02, NFR-PERF-03, NFR-SEC-03

5.3 See who's slipping and contact them (UC-03)

Use case	A coach finds their slipping members on Today and nudges one from their own phone
Actor	Coach. An owner does the same, for any member
Precondition	The coach is signed in on their phone, which has Messages or WhatsApp. One of their members is slipping.
Main flow	1. The coach opens Today. "Slipping now" lists their slipping members, the most days since the last visit first (FR-TODAY-01, FR-TODAY-02). 2. They press "Nudge →" on a member's row (FR-TODAY-02). 3. The composer, "Nudge {first}", opens. The coach picks a template, which fills in `{first}`, `{coach}`, `{gym}` and `{days}`, and edits the message as they like (FR-NUDGE-01, FR-TPL-02). 4. They press "Contact {first}". A menu offers "Text message", "WhatsApp" and "Copy message" (FR-NUDGE-02). 5. They pick "Text message". The same tap records the nudge as `text` and opens Messages with the member's number and the message filled in (FR-NUDGE-02, FR-NUDGE-03). 6. "Nudge to {first} recorded" shows. Open screens refresh, and Today's "Nudged this week" counts the nudge (FR-NUDGE-08, FR-TODAY-01, FR-LIVE-01). 7. The coach sends the message from Messages. Tenure sends nothing itself (FR-NUDGE-05). 8. The nudges log shows the nudge with its channel, "Text", and its outcome. It came back if the member checks in after it, within 7 days (FR-NUDGE-06, FR-NUDGE-07).

Use case	A coach finds their slipping members on Today and nudges one from their own phone
Alternatives	2a. The coach starts from "Nudge →" on the member page, or on the first slipping row of Today's board (FR-MEMBER-04, FR-BOARD-02). 2b. Another coach's member answers not found, and so does a member or a template from another gym (FR-STAFF-07, FR-NUDGE-08). 3a. The coach picks "Write my own" and writes the message from scratch (FR-NUDGE-01). 3b. Templates live at `/app/templates`. Owners create, edit and delete them, with a live preview; coaches read them (FR-TPL-01). 4a. On a computer the menu has no "Text message" (FR-NUDGE-02). 4b. The member has no phone: the composer says "No phone on file" and offers "Record nudge", which records the nudge as `in-person` (FR-NUDGE-04). 5a. "WhatsApp" records the nudge as `whatsapp` and opens WhatsApp with the message (FR-NUDGE-02, FR-NUDGE-03). 5b. "Copy message" copies the message and records the nudge as `copy`. The coach pastes it where they like (FR-NUDGE-02).
Postcondition	A nudge is stored with its member, sender, message, channel and time. The message waits in the coach's own app, or on the clipboard.
Requirements	FR-TODAY-01, FR-TODAY-02, FR-BOARD-02, FR-MEMBER-04, FR-NUDGE-01 to FR-NUDGE-08, FR-TPL-01, FR-TPL-02, FR-STAFF-07, FR-LIVE-01

5.4 Work the board and Members (UC-04)

Use case	Staff find members on the board and in Members, open one, and keep their details right
Actor	Owner or coach
Precondition	The user is signed in, and the gym has members.
Main flow	1. The user opens the board, from the navigation or from Today's link to it (FR-SHELL-01, FR-SHELL-02, FR-TODAY-05). 2. The board lists every member in their first 90 days, slipping first and then the newest, with 13 week marks, pace, status and the legend. A coach sees only their own members (FR-BOARD-01, FR-STAFF-07). 3. The user narrows the list with the filter bar: a search that folds case and accents, the status chips, and for an owner the coach filter. The filters stay in the URL (FR-MEM-02). 4. They press "Open" at the end of a row, named for the member, such as "Open Sam O." (FR-BOARD-02, FR-MEM-03). 5. The member page shows the member's facts, the days grid of their 90 days, their check-ins and their nudges. The phone shows in national format when its calling code matches the gym's country, and international otherwise (FR-MEMBER-01, FR-MEMBER-03). 6. An owner changes the member's coach, phone or name. Each change saves only its own field. Open screens in scope refresh, both coaches' when the coach changes (FR-MEMBER-02, FR-LIVE-01, FR-LIVE-02). 7. The user opens Members. It shows "First 90 days", and offers "Everyone" and "Archived". The same filter bar works there, names show as "First L." in the gym's locale order, and the list is a table from 48 rem and cards below (FR-MEM-01 to FR-MEM-04). 8. They press "Add member" and fill in "Add a member": a first name, an optional last name, a phone, a joined date of today or earlier, and a coach. Tenure adds the member (FR-MEM-05, FR-GLOBAL-04).

Use case	Staff find members on the board and in Members, open one, and keep their details right
Alternatives	2a. A check-in arrives while the board is open: its week cell fills with a pop. Nothing moves under reduced motion (FR-BOARD-03). 3a. Nothing matches: an empty state for the filters set. With no filter set and no members, the list says "Members are listed here once they join at the front desk or are imported." (FR-MEM-06). 4a. An unknown member, or another coach's, shows a not-found view (FR-MEMBER-01, FR-STAFF-07). 5a. "Nudge →" opens the composer (UC-03), and "Archive" starts UC-07 (FR-MEMBER-04, FR-MEMBER-05). 6a. A coach can't change these fields: the edit routes answer 403 (FR-STAFF-06). 6b. A blank first name, or a phone another member has, is refused, with the server's error under the field (FR-MEMBER-02). 8a. A coach adds a member: the new member is theirs (FR-MEM-05). 8b. A phone that's taken, or a joined date after today, is refused (FR-MEM-05). Any step: a screen whose data fails to load offers "Try again" (FR-SHELL-05).
Postcondition	The user has found and opened the members they wanted. Any edit or new member is stored, and open screens in scope show it.
Requirements	FR-SHELL-01, FR-SHELL-02, FR-SHELL-05, FR-TODAY-05, FR-BOARD-01 to FR-BOARD-03, FR-MEM-01 to FR-MEM-06, FR-MEMBER-01 to FR-MEMBER-05, FR-STAFF-06, FR-STAFF-07, FR-GLOBAL-04, FR-LIVE-01, FR-LIVE-02

5.5 Manage staff (UC-05)

Use case	An owner invites a coach or a co-owner, and later deactivates and reactivates staff
Actor	Owner, and the person invited
Precondition	The owner is signed in. The person invited has no account under their email.
Main flow	1. The owner opens Settings. Its Staff section lists each person with their role, their members in the 90-day window, how many are slipping, and "Inactive" where it applies (FR-STAFF-02). 2. They press "Invite staff", enter the person's name and email, and choose Coach or Owner (FR-STAFF-03). 3. They press "Create link". Tenure makes the invite and emails it to the outbox. The dialog says "We emailed this link to {email}. It works once and expires on {date}." and offers "Copy link" (FR-STAFF-03, FR-DEV-02). 4. The owner copies the link and gives it to the person, since no email leaves the machine. In stall mode the link uses the laptop's LAN address (FR-STAFF-03). 5. The person opens the link. "Join {gymName}" shows, with their name filled in from the invite, and the owner wording on an owner invite (FR-STAFF-04). 6. They keep or change their name, choose a password, and press "Join {gymName}". Tenure creates their account in the gym with the invite's role, uses up the invite, and signs them in. Any old session in that browser ends (FR-STAFF-01, FR-STAFF-04). Which email the account gets is still open (Appendix B, OI-15). 7. The staff list shows the new person (FR-LIVE-01). 8. Later the owner presses "Deactivate" on the person's row. The confirm, "Deactivate {name}", says they won't be able to log in and their members stay on the board. Once it's confirmed, the person is refused at log-in, and their sessions and live stream end at once. Their history stays, and the row shows "Inactive" (FR-STAFF-05, FR-AUTH-06, FR-LIVE-04). 9. The owner presses "Reactivate", and the person can log in again (FR-STAFF-05).

Use case	An owner invites a coach or a co-owner, and later deactivates and reactivates staff
Alternatives	2a. A missing name or role is refused (FR-STAFF-03). 3a. In development the email can be read at `/dev/outbox` (FR-DEV-01). 5a. A bad or used link, or one older than 7 days, shows "This link doesn't work" (FR-STAFF-03, FR-STAFF-04). 6a. The email already has an account: refused, and the invite stays unused (§6.1, `invites`). 6b. An owner invite makes a co-owner, with every owner right (FR-STAFF-01, FR-STAFF-06). 8a. The person is the last active owner: refused with "The last active owner can't be deactivated." (FR-STAFF-05). Any step: a coach has no Settings in their navigation, and its routes answer 403 (FR-STAFF-06, FR-SHELL-04).
Postcondition	The person has an active account in the gym with the invite's role, and the invite can't be used again. A new coach sees only their own members (FR-STAFF-07). After step 8 they're inactive, with their history kept; after step 9, active again.
Requirements	FR-STAFF-01 to FR-STAFF-07, FR-AUTH-06, FR-DEV-01, FR-DEV-02, FR-SHELL-04, FR-LIVE-01, FR-LIVE-04

5.6 Read Reports (UC-06)

Use case	An owner sees how the gym's new members are doing, and what nudges achieve
Actor	Owner
Precondition	The owner is signed in, and the gym has members with some history.
Main flow	1. The owner opens Reports, a page only owners have (FR-REPORT-01, FR-SHELL-04). 2. The cohorts show each join month's members, and the share still coming at day 30, 60 and 90. Months follow the gym's locale (FR-REPORT-01, FR-REPORT-03). 3. The visits-by-week chart draws still coming as a solid line in the text colour, slipped as a dashed line in the slipping colour with point markers, and the 3-a-week line dotted in soft text. Each line is labelled at its end, with no legend, and screen readers get a description (FR-REPORT-02). 4. The nudge figures show how many nudges were sent, how many came back, and the rate. Nudges are counted in the gym's time zone (FR-REPORT-01, FR-REPORT-03, FR-NUDGE-07). 5. Archived members' history counts in every figure. A deleted member's no longer does (FR-REPORT-01, FR-MEMBER-07, ARCHITECTURE §2).
Alternatives	1a. A coach has no Reports in their navigation, and its route answers 403 (FR-STAFF-06, FR-SHELL-04). 1b. The data fails to load: the page offers "Try again" (FR-SHELL-05). 1c. The owner opens Billing instead. It shows "Pilot day {n} of 90", the 90 days as one bar, and that the pilot is free, with no price (FR-BILL-01). The same pilot day shows by the gym name from 768 px (FR-SHELL-03).
Postcondition	Nothing changes: Reports and Billing only read.
Requirements	FR-REPORT-01 to FR-REPORT-03, FR-NUDGE-07, FR-MEMBER-07, FR-BILL-01, FR-STAFF-06, FR-SHELL-03 to FR-SHELL-05

5.7 Archive, restore and delete a member (UC-07)

Use case	Staff archive a member who has left, or bring them back; an owner deletes one for good
Actor	Owner, or a coach for their own members
Precondition	The user is signed in and can see the member (FR-STAFF-07).

Use case	Staff archive a member who has left, or bring them back; an owner deletes one for good
Main flow	1. On the member page the user presses "Archive" (FR-MEMBER-05). 2. The confirm, "Archive this member?", says they leave every list and the kiosk search, and that their history stays. The user presses "Archive member" (FR-MEMBER-05). 3. Tenure stores when the member was archived. They leave Today, the board, Members' "First 90 days" and "Everyone" views and the kiosk's name search, and show in "Archived" (FR-MEM-01, FR-KIOSK-02). 4. Open screens refresh. Reports still count the member's history (FR-LIVE-01, FR-REPORT-01).
Alternatives	1a. A coach archives only their own members. Another coach's member answers not found (FR-MEMBER-05, FR-STAFF-07). 4a. Restore: the user opens the member from "Archived" and presses "Restore member", with the same permissions as archiving. The member is back in every list, with their history (FR-MEMBER-06, FR-LIVE-01). 4b. Delete: on the archived member's page an owner presses "Delete permanently". The confirm, "Delete this member for good?", says their visits and nudges go too. The owner presses "Delete member". Tenure deletes the member with their check-ins and nudges, and the screen goes back to Members. Reports lose them, and older backups still hold them (FR-MEMBER-07, FR-PRIV-03, NFR-PRIV-02). 4c. The server refuses to delete a member who isn't archived: "Archive this member before deleting them." (FR-MEMBER-07). 4d. The member turns up at the kiosk: their phone, or a join with it, shows "Please see the front desk to check in." Staff restore them (4a), and they check in (FR-KIOSK-06). 4e. An import row that matches the member restores them, and the check step says so (FR-IMPORT-09).
Postcondition	The member is archived, with their history kept and counted; or restored to every list; or deleted, with their check-ins and nudges, from the live database.
Requirements	FR-MEMBER-05 to FR-MEMBER-07, FR-MEM-01, FR-KIOSK-02, FR-KIOSK-06, FR-IMPORT-09, FR-REPORT-01, FR-PRIV-03, FR-STAFF-07, FR-LIVE-01, NFR-PRIV-02

5.8 Edit Settings (UC-08)

Use case	An owner sets the gym's name, country and time zone, and opens Tenure on the kiosk tablet and on phones
Actor	Owner
Precondition	The owner is signed in. For the tablet and phones, Tenure runs in stall mode and they are on the laptop's network (FR-OPS-01).
Main flow	1. The owner opens Settings (FR-GYM-01). 2. They change the gym's name, its country (one of the five) or its time zone. A US gym's zone list holds the old app's 7 zones, plus the current one if it's missing. Other countries list all of their zones by city, east to west (FR-GYM-01, FR-GYM-02, FR-GLOBAL-01). 3. They save. "Settings saved" shows, and the gym's details update everywhere, open screens included (FR-GYM-01, FR-GYM-05). 4. From then on, phones are read in the new country, and dates and times follow its locale and the zone. Stored phones don't change, since they are E.164 (FR-GLOBAL-02, FR-GLOBAL-03). 5. In the Kiosk section, the owner opens the kiosk link on the front-desk tablet, by its QR code or by copying it. "Open the kiosk" opens it in a new tab (FR-GYM-03). 6. In "Open on a phone", the owner scans the QR code with a phone, or copies the link. In stall mode both links use the laptop's LAN address (FR-GYM-03).

Use case	An owner sets the gym's name, country and time zone, and opens Tenure on the kiosk tablet and on phones
Alternatives	2a. No country, or one outside the five: refused with "Choose a country from the list." (FR-GYM-01). 2b. A zone that isn't on the list is refused (FR-GYM-02). 6a. No LAN address is found: the phone section says "No network address found. Connect this computer to Wi-Fi, then reload the page." (FR-GYM-03). 6b. The Import box's "Import members or check-ins" opens the import page (FR-GYM-04, UC-01). The Staff section is UC-05. Any step: a coach has no Settings, and its routes answer 403 (FR-STAFF-06).
Postcondition	The gym's name, country and zone are saved and shown everywhere. The kiosk runs on the tablet, and the owner's phone has the app.
Requirements	FR-GYM-01 to FR-GYM-05, FR-GLOBAL-01 to FR-GLOBAL-03, FR-STAFF-06, FR-OPS-01

5.9 Reset a password (UC-09)

Use case	A staff member who forgot their password chooses a new one
Actor	Staff member; at the stall, the operator
Precondition	The staff member has an account, and isn't signed in.
Main flow	1. On Login they press "Forgot your password?" (FR-AUTH-07). 2. "Reset your password" asks for their email, and they press "Send reset link" (FR-AUTH-07). 3. Tenure answers the same whether or not the email has an account: "If an account uses that email, a reset link is on its way. It works once, for an hour." When it has one, the link is emailed to the outbox (FR-AUTH-07, FR-DEV-02). 4. They get the link. In development it's on `/dev/outbox`. On the stall laptop and in Docker that page is off, so the operator runs `npm run reset-link -- <email>`, which prints a link in the terminal (FR-DEV-01, FR-AUTH-08, NFR-SEC-06). 5. The link opens "Choose a new password". They enter a new password and press "Save password" (FR-AUTH-07). 6. Tenure saves it and uses up the link. "Password saved" shows, and the user's other sessions end (FR-AUTH-07).
Alternatives	2a. Too many reset requests: "Too many reset requests. Try again in 10 minutes." (NFR-SEC-03). 3a. No account has that email: the same answer, and nothing is emailed (FR-AUTH-07). 5a. The link is unknown, used, or over an hour old: "This link doesn't work", and the server's sentence says to get a new one from the log-in page (FR-AUTH-07). 5b. A new password shorter than 8 characters is refused, as the field's help ("8 characters or more.") says.
Postcondition	The user can log in with the new password, and not with the old one (FR-AUTH-04). The link can't be used again, and the user's other sessions have ended.
Requirements	FR-AUTH-04, FR-AUTH-07, FR-AUTH-08, FR-DEV-01, FR-DEV-02, NFR-SEC-03, NFR-SEC-06

5.10 Run the stall demo (UC-10)

Use case	Visitors at the expo stall try Tenure on the demo gym, and the team resets it between them
Actor	Stall visitor; the operator
Precondition	The demo gym was seeded on the morning of the stall, and the operator started stall mode: `npm run stall`, or Docker through `scripts/stall.sh`. The kiosk tablet and the team's phones are on the laptop's network (FR-OPS-01).

Use case	Visitors at the expo stall try Tenure on the demo gym, and the team resets it between them
Main flow	1. The demo gym, Northside Strength, took the laptop's country from its time zone when it was seeded: the US outside the five, unless `TENURE_COUNTRY` overrides it. Its sample numbers stay +1 (FR-GLOBAL-05). 2. The tablet shows the demo gym's kiosk, with the "Sample data" tag (FR-KIOSK-07). 3. A visitor joins at the kiosk with a first name, and is checked in (UC-02). 4. The visitor taps "Try the demo", and is signed in as the demo gym's owner, with everything usable (FR-WEB-04). 5. They find themselves in "Checked in today", then try the board, a nudge, an archive or a delete (FR-TODAY-01; UC-03, UC-04, UC-07). 6. When the visitor leaves, the operator runs `npm run demo:reset`, or presses the stall-mode button, and the demo gym is re-seeded (FR-OPS-03). 7. The next visitor starts from a fresh demo gym.
Alternatives	4a. Outside stall mode there is no "Try the demo" (FR-WEB-04). 6a. Until the reset, the next visitor sees what earlier visitors entered, names and phones included (§2.6). 6b. The operator backs up with `npm run backup`, or `scripts/backup.sh` for the Docker volume. `npm run restore -- <file>` puts a backup back, and is refused while the server runs or when the file fails its integrity check (FR-OPS-02). 6c. What visitors entered stays in any backup taken before the reset (NFR-PRIV-02). Any step: `docs/DEMO.md`, the stall's runbook, has these steps (FR-OPS-05).
Postcondition	The demo gym is freshly seeded, and what visitors entered is gone from the live database.
Requirements	FR-OPS-01 to FR-OPS-03, FR-OPS-05, FR-WEB-04, FR-GLOBAL-05, FR-KIOSK-07, FR-TODAY-01, NFR-PRIV-02

6. Data requirements

All of Tenure's data is one SQLite file on the machine that runs it (FR-PRIV-01). The schema is in `packages/db/src/schema`, and only `packages/db` talks to the database (ARCHITECTURE §4). Migrations run on start, so an older database upgrades (FR-OPS-04, NFR-REL-02):

Migration	Adds
0000	The 8 tables: `gyms`, `users`, `sessions`, `invites`, `members`, `checkins`, `templates` and `nudges`
0001	`gyms.country`
0002	`members.archived_at`
0003	`password_resets` and `outbox`

6.1 Entities

The 0000 tables are keyed by an integer `id`, except `sessions` and `invites`, which are keyed by their token's hash.

Table	Key fields	Rules
`gyms`	`name`, `slug`, `timezone`, `country`, `is_demo`, `pilot_started_on`, `plan`, `kiosk_token`, `created_at`	▪ One row per gym: the tenant the app data hangs off. ▪ `slug` and `kiosk_token` are unique. A taken slug gets `-2`, `-3` and so on. ▪ `kiosk_token` is the kiosk link's secret. It is stored as it is, since Settings shows the link (FR-KIOSK-01, FR-GYM-03). ▪ `country` is an ISO 3166-1 alpha-2 code: `US`, `CA`, `GB`, `PK` or `AE` (FR-GLOBAL-01). Migration 0001 gives an older gym `US`. ▪ `timezone` is an IANA zone, such as `America/Chicago`: the browser's at sign-up, and changeable in Settings (FR-AUTH-01, FR-GYM-01). ▪ `pilot_started_on` is the pilot's first day, which Billing counts from (FR-BILL-01). `plan` is `pilot` for every gym. ▪ `is_demo` marks the demo gym (FR-KIOSK-07, FR-OPS-03). ▪ No screen deletes a gym.
`users`	`gym_id`, `email`, `name`, `role`, `password_hash`, `active`, `created_at`	▪ `email` is unique across every gym on the machine, ignoring case: one account per email. ▪ `role` is `owner` or `coach`, held by a CHECK (FR-STAFF-01). ▪ `password_hash` is scrypt (NFR-SEC-01). ▪ `active` is false while the person is deactivated. Deactivating deletes all of their sessions (FR-AUTH-06, FR-STAFF-05). ▪ No screen deletes a user: staff are deactivated, and keep their history (FR-STAFF-05).

Table	Key fields	Rules
`sessions`	`token_hash`, `user_id`, `created_at`, `expires_at`	▪ The cookie holds a random token. Only its sha256 is stored, as the key (NFR-SEC-01). ▪ `expires_at` is 30 days after `created_at` (FR-AUTH-05). ▪ Log-out deletes the session, and an expired one is deleted when it's next presented. A password reset ends the user's other sessions (FR-AUTH-05, FR-AUTH-07). ▪ Sessions cascade from their user.
`invites`	`token_hash`, `gym_id`, `role`, `name`, `created_by`, `created_at`, `used_at`	▪ Only the sha256 of the link's token is stored, as the key. ▪ `role` is `coach` or `owner`, held by a CHECK, and the new user gets it (FR-STAFF-03, FR-STAFF-04). ▪ `name` fills the invite page's name field. `created_by` is the owner who made the invite. ▪ An invite is live while `used_at` is null and it's under 7 days old. Accepting sets `used_at` and creates the user, in one transaction. An email that already has an account is refused, and the invite stays unused (FR-STAFF-03, FR-STAFF-04). ▪ The 0000 table keeps no email. The invitee sets only a name and a password (FR-STAFF-04), so the new account needs the address the invite was sent to. Proposed: a column for it in migration 0003 (Appendix B, OI-15).
`members`	`gym_id`, `first_name`, `last_name`, `phone`, `email`, `joined_on`, `coach_id`, `external_id`, `source`, `created_at`, `archived_at`	▪ `first_name` is required. `last_name` may be null, for a single name (FR-GLOBAL-04). ▪ `phone` is E.164 (FR-GLOBAL-02). It and `external_id`, the member's ID in the gym's own system, are each unique within the gym. Blank text is stored as null, never `''`, so many members may have neither. ▪ Archiving keeps the phone, so it stays taken: a join with it shows the front-desk line (FR-KIOSK-06). ▪ `joined_on` is a date in the gym's time zone: today for a kiosk join, today or earlier for "Add member", and from the file for the import (FR-KIOSK-04, FR-MEM-05, FR-IMPORT-02). ▪ `coach_id` is one of the gym's staff, or null for "No coach" (FR-MEMBER-02). ▪ `email` comes only from the import and the seed. ▪ `source` says how the member came in: `kiosk`, `csv` or `seed`, plus one value for members added by hand (FR-MEM-05). ▪ `archived_at` is null for an active member, and the time of archiving otherwise. "Restore member" and a matching import row clear it (FR-MEMBER-05, FR-MEMBER-06, FR-IMPORT-09). ▪ Only an archived member can be deleted, and their check-ins and nudges cascade (FR-MEMBER-07).

Table	Key fields	Rules
`checkins`	`gym_id`, `member_id`, `at`, `source`	▪ `at` is when the member arrived. `source` is `kiosk`, `csv` or `seed`. ▪ A second check-in by the same member within an hour isn't stored, at the kiosk or by the import. The writes check this; no index does (FR-KIOSK-03, FR-IMPORT-07). ▪ The import also skips a check-in in the future, or from before the member joined (FR-IMPORT-07). ▪ Check-ins cascade from their member.
`templates`	`gym_id`, `name`, `body`, `sort`, `updated_at`	▪ `name` is up to 60 characters and `body` up to 1,000 (FR-TPL-01). ▪ `sort` orders the list, and a new template goes last. ▪ A new gym gets the three default templates (FR-TPL-03). ▪ Deleting a template keeps the nudges made from it, with their text. Their `template_id` becomes null.
`nudges`	`gym_id`, `member_id`, `user_id`, `template_id`, `body`, `channel`, `sent_at`	▪ `user_id` is the sender. `template_id` is the template it started from, or null. ▪ `channel` is `text`, `whatsapp`, `copy` or `in-person` (FR-NUDGE-02, FR-NUDGE-04). `own-phone`, the column's default, marks a nudge from before the first version, which shows no channel (FR-NUDGE-06). The column is plain text, and the code's types keep other values out. ▪ The member and the template must belong to the nudge's gym (FR-NUDGE-08). ▪ Whether a nudge came back isn't stored: it's worked out from check-ins (FR-NUDGE-07). ▪ Nudges cascade from their member.
`password_resets`	Added by migration 0003, with password reset; its columns are set then. It holds the user each link is for, the link's secret, and what makes the link work once and expire.	▪ A link works once, and expires after an hour (FR-AUTH-07). ▪ "Send reset link" makes one only for an email with an account, though the answer is the same either way. `npm run reset-link` makes one too (FR-AUTH-07, FR-AUTH-08). ▪ Saving a new password with it ends the user's other sessions (FR-AUTH-07).
`outbox`	Added by migration 0003. One row per email the email stand-in sends: the recipient, the subject and the text, as the email adapter receives them.	▪ Invite and reset emails land here, and no email leaves the machine (FR-STAFF-03, FR-AUTH-07, FR-DEV-02). ▪ Its links work, so `/dev/outbox` shows it only in development. Elsewhere the operator prints reset links in the terminal (FR-DEV-01, FR-AUTH-08, NFR-SEC-06).

6.2 Rules across the tables

Rule	What it means
One gym	Users, invites, members, check-ins, templates and nudges each carry a `gym_id`, and sessions belong to a user. Every read and write is held to the user's own gym (FR-PRIV-02). SQLite enforces the foreign keys.

Rule	What it means
On delete	`ON DELETE CASCADE` runs from a gym to every row with its `gym_id`, from a member to their check-ins and nudges, and from a user to their sessions. `members.coach_id` and `nudges.template_id` are set to null instead. `nudges.user_id` and `invites.created_by` have no action, so a user who sent a nudge or made an invite can't be deleted: staff are deactivated instead (FR-STAFF-05).
Dates	`YYYY-MM-DD` text, in the gym's time zone: `joined_on` and `pilot_started_on`.
Timestamps	UTC ISO 8601 text, such as `2026-11-06T14:00:00.000Z`: `created_at`, `updated_at`, `expires_at`, `used_at`, `archived_at`, `checkins.at` and `nudges.sent_at`. The repositories hide the format, so a move to Postgres changes only `packages/db` (ARCHITECTURE §6).
Phones	Stored only on members, always as E.164, such as `+13125550142`. One rule on the server reads every phone, in the gym's country (FR-GLOBAL-02).
Countries	`gyms.country` only: `US`, `CA`, `GB`, `PK` or `AE` (FR-GLOBAL-01).
Channels	`nudges.channel` only: `text`, `whatsapp`, `copy` and `in-person`, plus `own-phone` for nudges from before the first version (FR-NUDGE-06).
Tokens	Session and invite tokens are stored as sha256 hashes (NFR-SEC-01). The kiosk token is stored as it is, since it is the kiosk link.
Uniqueness	`gyms.slug`; `gyms.kiosk_token`; `users.email`, ignoring case; and on members, `(gym_id, phone)` and `(gym_id, external_id)`. The one check-in an hour is a rule in the writes, not a constraint.

6.3 Retention and deletion

Data	How long it's kept
Members, with their check-ins and nudges	Until an owner deletes the member (FR-MEMBER-07). Archiving keeps them, and Reports still count them (FR-REPORT-01).
Staff	Kept: no screen deletes a user, and deactivated staff keep their history (FR-STAFF-05).
Templates	Until an owner deletes one. Nudges made from it keep their text (FR-TPL-01).
Sessions	30 days at most (FR-AUTH-05).
Invites and reset links	Usable once: an invite for 7 days, a reset link for an hour (FR-STAFF-03, FR-AUTH-07). No limit is set on keeping spent ones.
Outbox emails	No limit is set.
The demo gym	Re-seeded between stall visitors, which clears what they entered from the live database (FR-OPS-03).
Backups	Each `npm run backup` writes a dated copy of the whole database to `backups/` (FR-OPS-02). The operator keeps or deletes them.

Deleting a member for a data-protection request (FR-PRIV-03):

- The member, their check-ins and their nudges leave the live database at once, and Reports lose them (FR-MEMBER-07, ARCHITECTURE §2).
- Older backups still hold them, and DEMO.md says so (NFR-PRIV-02). Restoring a backup taken before the delete brings them back.

7. Architecture and key algorithms

7.1 The monorepo

Tenure is one TypeScript monorepo, on npm workspaces and Turborepo (ARCHITECTURE §3). Each part imports only the parts its last column lists, and ESLint's boundary rules hold it to that (NFR-MAINT-01).

Part	What it holds	Imports from the repo
`apps/api`	The Hono API on Node 26. In stall mode and Docker it also serves the built web app	`db`, `domain`, `contracts`, `copy`
`apps/web`	The React SPA: the website, the app under `/app`, the kiosk, and the invite and reset pages	`ui`, `copy`, `contracts`, `domain`
`apps/e2e`	Playwright's flows, each browser against its own seeded stall-mode server	Outside the rules: `copy`, `domain`, and the seed in `db` for the demo logins
`packages/contracts`	zod schemas for every request, response, live event and error code: the API's single source of truth	`domain`
`packages/domain`	Every business rule, pure: dates in the gym's time zone, weeks, pace, status, KPIs, the orders, came back, templates, phones, the report maths and the CSV pipeline	Nothing
`packages/db`	The Drizzle schema, migrations, client, repositories and seed. Nothing else talks to the database	Nothing
`packages/ui`	The design tokens for both themes, their CSS variables, the theme runtime, primitives and icons	Nothing
`packages/copy`	COPY.md, generated into a typed catalogue, so `t('id')` fails to compile for a missing id	Nothing
`packages/config`	The shared tsconfig, ESLint and Vitest presets	Nothing

The outside libraries stay where the contract puts them: libphonenumber-js in `domain`, zod in `contracts`, Drizzle in `db`, and React and Radix in `ui` (ARCHITECTURE §4). Apps never import another app, and packages never import an app. The domain gets the clock and the time zone as arguments: ESLint bans Node modules, `process`, `Date.now()` and an argument-less `new Date()` there. `marketing/` (the expo kit) and `legacy/app/` (the old app, the port's reference) don't change during the port.

An API module is a folder, `apps/api/src/modules/<name>/`, plus one line in `modules/index.ts` (ARCHITECTURE §5):

- `index.ts` exports its `ApiModule`: a name, `register(router, ctx)`, an optional `subscribe(bus, ctx)`, and the routes it declares public or owner only. Every other route needs a signed-in user (NFR-SEC-04).
- `routes.ts` validates input against the contract, calls the service and returns contract-shaped JSON. An error is `{ error: { code, message } }`, with its message from copy.
- `service.ts` holds the use cases. It calls repositories and domain rules, and after a successful write it publishes a domain event, such as `checkin.recorded` (FR-LIVE-01). Routes never publish.
- A module reaches another only through that module's `index.ts`, or its events. `ctx` carries the rest: the repositories, the event bus, the clock, config, the adapters, the rate limits, shutdown and the LAN addresses. Tests build it with an in-memory database, a fixed clock and a fixed address.

- The `realtime` module sends events down one server-sent stream per tab, `GET /api/events`, in each user's scope (FR-LIVE-02).

A web feature is a folder, `apps/web/src/features/<name>/`, plus one line in `app/features.ts` (ARCHITECTURE §8):

- `index.ts` exports its `WebFeature`: its routes, lazy-loaded so the website ships without the app (FR-WEB-05); its nav items; and which of its queries each live event makes stale.
- `screens/`, `components/`, and `api.ts` with its query keys and hooks. Server state lives in TanStack Query.
- A feature imports only `shared/` and packages, and links to other features by URL. Only `app/` knows every feature.
- `shared/live` holds the tab's one EventSource and turns its events into batched refreshes (FR-LIVE-03).

The data stays in SQLite behind the repositories, with dates as ISO text, so a move to Postgres changes only `packages/db` (§6). The design system names its tokens by role, never by colour. Tests check every colour pair (NFR-USE-01) and fail on a literal colour, duration, z-index or opacity in component CSS (ARCHITECTURE §9).

7.2 Key algorithms

Each rule is written once, in `packages/domain`, and keeps its tests (ARCHITECTURE §7). Dates are `YYYY-MM-DD` in the gym's time zone, and the clock and the time zone are always passed in.

7.2.1 Weeks, pace and status

The terms are defined in §1.3; Today and the board show the results (FR-TODAY-01, FR-BOARD-01).

1. A visit is a local day, in the gym's time zone, with at least one check-in. Several check-ins on one day count once.
2. Day 0 is the joined date. Only visits on days 0 to 89, up to today, count. Week n holds days 7(n−1) to 7n−1, and week 13, cut short by the window, holds days 84 to 89.
3. Week n is finished once today is day 7n or later, and week 13 once today is day 90. The week that holds today is in progress.
4. Pace is the mean visits of the last 3 finished weeks, or of all of them while fewer have finished, to one decimal. There is none until week 1 finishes.
5. Status takes the first rule that matches, over the finished weeks:
    1. slipping: 2 or more, and the last two each under 2 visits;
    2. new: fewer than 2;
    3. on the line: the last one had 3 or more visits;
    4. below the line: anything else.
6. Today's four KPIs count the viewer's members in their first 90 days (FR-TODAY-01):
    - new members: how many there are;
    - the share on the line: the percent on the line among those with at least one finished week, or none when no one has one;
    - slipping: how many are slipping;
    - nudged this week: nudges sent in the last 7 days, the whole gym's for an owner, and the ones a coach sent for a coach.

For example, on day 20 a member has two finished weeks. With 4 visits in week 1 and 1 in week 2, they are below the line, at a pace of 2.5. With 1 visit in each, they are slipping.

7.2.2 Came back

1. A nudge came back when the member has a check-in strictly after the nudge was recorded and no more than 7 days (7 × 24 hours) after it (FR-NUDGE-07). A check-in at the same instant doesn't count, and one at exactly 7 days does.
2. The rule compares instants, so the gym's time zone doesn't change the answer.
3. It is worked out from the check-ins each time it's shown, and never stored (§6.1).
4. The nudges log shows the outcome (FR-NUDGE-06): "Came back"; otherwise "No visit yet" while the nudge is under 7 days old, and "No visit in 7 days" after that.
5. Reports count how many of the last 90 days' nudges came back (§7.2.7).

7.2.3 Board order and "Slipping now" order

The board (FR-BOARD-01):

1. It lists the viewer's members in their first 90 days: the whole gym for an owner, and their own members for a coach (FR-STAFF-07). Archived members are left out (§1.3).
2. Slipping members come first.
3. Within each group, the newest joiners come first, by joined date, then the lower id, so the order is stable.
4. Today's board keeps this order, so its first slipping row, the one with "Nudge →", is its first row whenever anyone is slipping (FR-BOARD-02).

"Slipping now" on Today (FR-TODAY-02):

1. It lists the slipping members of the same list.
2. The viewer's own members come first: those they coach, so an owner who also coaches sees theirs at the top.
3. Then the most days since the last visit come first. A member who never visited counts as one day more than their days since joining.
4. Then the lower id.

7.2.4 The phone rule: the gym's country wins

1. A trailing extension, such as "x12" or "ext. 12", is dropped, and the digits are kept. With no digits there is no phone.
2. The digits are read in the gym's country, where a trunk 0, the country code and the country's own international prefix (011 in the US, 00 in the UK) all work. If they make a real number there, that is the answer, with or without a "+" (FR-GLOBAL-02, answers.md Q34).
3. Otherwise a number without "+" is refused.
4. A "+" number is read as international, and kept if it is a real number for its country code.
5. Failing that, it is kept if its country isn't the gym's and its length is possible there, since the phone data can lag behind newly assigned numbers. The gym's own code is never kept on length alone.
6. What is kept is stored as E.164. Only the server runs this rule, on libphonenumber's full metadata, for the kiosk, Add member, the member page and the import. The web only formats numbers (FR-MEMBER-03), and ESLint keeps the rule out of it.

Gym	Typed	Stored	Why
US	+312 555 0142	+13125550142	A real Chicago number wins over the Netherlands (+31)
US	011 44 20 7946 0958	+442079460958	The US's own international prefix
US	+1 555 555 0142	Refused	555 is not an assigned area code, and the gym's own code needs a real number
US	+290 2345	+2902345	A foreign number needs only a possible length
UK	07911 123456	+447911123456	A local mobile
UK	(312) 555-0142	Refused	Not a UK number, and it has no "+"
Pakistan	3125550142	+923125550142	A real Pakistani mobile, so the demo's +1 numbers are typed with "+1" there

The trade-off, which Uzair accepted: a real foreign number whose digits also make a local one is stored as local.

7.2.5 The import's date order

1. The values come from the file's date column: the join dates for members, or the check-in times for check-ins.
2. A year-first date, such as 2026-03-04, reads one way only and settles nothing.
3. In any other date, a first number over 12 can only be a day, so that date reads only day first. A second number over 12 means it reads only month first.
4. If some dates read only one way and none only the other, the file settles the order. The line says "Set from the file: some of its dates only read this way."
5. If no date settles it, or dates point both ways, the gym's country sets it: the order its locale writes dates in (FR-GLOBAL-03), such as month first in the US and day first in the UK. The line says "Set from your gym's country. Check the first rows below."
6. The owner's flip, "Read them day first" or "Read them month first", wins over both. The first rows show as read in the order chosen, such as "Row 2: 03/04/2026 is 3 Apr 2026", or say they can't be read that way (FR-IMPORT-04).

7.2.6 Import matching and planning

The planner is pure. It returns what to create, update and skip, with problem codes that the API words from copy. The check step and the commit run the same plan, and only the commit writes (FR-IMPORT-08).

1. **Reading.** The file is read as CSV, and its first non-blank line is the header. Rows keep their spreadsheet numbers, the header being row 1, so each "Row {n}: ..." points at the right line (FR-IMPORT-05). A row with more or fewer columns than the header is reported.
2. **Mapping.** Each field's column is suggested by matching the headers, in lower case with only letters and digits kept, against that field's common export names, in priority order. A column feeds one field at most, and the owner can change any choice. Without the required columns nothing is planned, and the page names what's missing (FR-IMPORT-02).
3. **A member row** is read field by field: the name, splitting a "Full name" (FR-IMPORT-03); the join date, in the date order (§7.2.5); the phone, by the phone rule (§7.2.4); the email; and the coach, matched by email to the gym's staff. A row with no name, or no readable join date, is skipped. A bad phone or email is left blank, and an unknown coach leaves no coach, each with a note (FR-IMPORT-05).

4. **Matching.** The row is matched to the gym's members, archived ones included: by external ID first, then by phone (FR-IMPORT-06).
5. **A matched row** is skipped when an earlier row matched the same member, or when it would give them an ID an earlier row used. Otherwise it updates only the fields that differ, and a blank never overwrites. With nothing to change, it is skipped as up to date. A phone that is another member's is left as it was, with a note. An archived member is restored, and the row says so (FR-IMPORT-09).
6. **An unmatched row** is skipped when its external ID or phone came earlier in the file. Otherwise it creates a member.
7. **A check-in row** finds its member by ID, then by phone, read by the phone rule. It is skipped when no member matches, when its time can't be read in the gym's time zone and date order, or when the time is in the future, on a day before the member joined, or within an hour of another of that member's check-ins, stored or earlier in the file (FR-IMPORT-07).
8. **The commit** plans again inside one write transaction, against the gym's data as it is then, so a kiosk check-in made in the meantime counts for the hour rule. It writes the creates, updates and check-ins together, or nothing on a clash. A row's coach and a check-in's member must belong to the gym (FR-IMPORT-08).
9. **The result** shows what was created, updated and skipped, and open screens refresh once (FR-IMPORT-10, FR-LIVE-03). The check step for 2,000 rows shows in under 5 s (NFR-PERF-04).

7.2.7 Report cohorts and the visits-by-week chart

Reports count every member who hasn't been deleted, archived ones included (FR-REPORT-01). Visits are days in the gym's time zone (FR-REPORT-03).

Cohorts:

1. The rows are the gym's current month and the three before it, oldest first. Each holds the members who joined that month.
2. For day N, which is 30, 60 or 90, a row takes its members who are at least N days in.
3. Its figure is the percent of them who visited in the week before day N, days N−7 to N−1, as a whole number. With no member that far in, there is no figure.

For example, a member who joined on 1 August counts toward day 30 from 31 August, and is still coming if they visited between 24 and 30 August.

Visits by week:

1. For each of weeks 1 to 13, the chart takes every member whose week has finished.
2. It splits them by their last visit: still coming if it was in the last 14 days, today included, and slipped otherwise, which includes members who never visited.
3. Each line's point is its group's mean visits that week, to one decimal. An empty group has no point.
4. The line at 3 visits a week is drawn with them (FR-REPORT-02).

The nudge figures count the nudges sent in the last 90 days, by the gym's day and up to now; how many came back (§7.2.2); and the rate, as a whole percent, or none when nothing was sent.

8. Build plan

The first version is built in six phases, about 40 jobs in all, from `docs/milestone-2/FIRST-VERSION.md`. They take about 14 to 15 hours, plus the time Uzair's answers take. `docs/PROGRESS.md` says where the build stands.

How the jobs run:

- A job is one area through every layer: db, contracts, api and web, with its tests. One agent builds it, in its own git worktree off `m2-core`, with 6 agents at a time.
- Registration is written once, in phase 0, so later jobs edit only their own folders. Each shared file has one owner per phase: the copy, the errors, the index files, the schema and migrations, `packages/ui`, the root package files and the e2e fixtures. Where two jobs in a phase meet, a typed stub comes first.
- Tests are named by the requirement ID they cover (NFR-MAINT-03).
- Review: "Fresh" below means a fresh reviewer for the job, for the 17 jobs that touch sign-in, permissions, data writes, the import, the kiosk, deletion or email. "Phase" means the job's tests, plus one reviewer reading the phase's merged diff.
- A job gets at most two repair rounds. One that fails twice goes to a fresh builder.
- Merges are `--no-ff` into `m2-core`, in the order copy, db, domain and contracts, api, ui, web. After each phase's merges, a runner runs the five gates (NFR-MAINT-02). Nothing is pushed.
- Product questions stop for Uzair, and the other jobs carry on.

8.1 Phase 0: foundations (about 2 h, while the design picks are made)

First the orchestrator makes the contract commits, one change each: to ARCHITECTURE §2, §6, §8, §11, §12 and §13, and the rule that Playwright is the one test-only package. Then it writes this spec, §3's IDs first.

Job	Builds	Requirements	Review
C0	Copy v1.7 with every decided word, every first-version error code, and the contract placeholders	NFR-COPY-01	Phase
X0	Migrations 0001 (`gyms.country`) and 0002 (`members.archived_at`); the nudge channel enum; a country for sign-up and settings, and a role for invites; the seed's country and channels; `AuthGym.country`; the northside and demo-login ports	FR-GLOBAL-01, FR-GLOBAL-05, FR-NUDGE-02, FR-OPS-04, NFR-REL-02	Fresh
PL0	`scopeOf` in platform, `idParam`, `readQuery` and `ctx.lan`; every API module and web feature registered once as a stub; `patch` and `del` in the client; live refreshes batched per tick; PageTitle	FR-LIVE-03, FR-SHELL-06, NFR-SEC-04	Fresh
G0	`normalizePhone(input, country)`, banned from the web by ESLint; `localeFor`, with en-001 and each country's clock; the five markets and their zones; `formatPhone`; the kiosk's as-you-type number; `fold()`	FR-GLOBAL-01 to FR-GLOBAL-03, FR-GYM-02, FR-MEMBER-03, FR-KIOSK-02, FR-MEM-02, NFR-LOC-01	Phase
UI1	ChoiceGroup, the marks, DaysGrid, ConfirmDialog, FormError, Table, FactList, CopyField, FileButton, DayTrack and the lg Button; the N1 kiosk blend, `line-blend`, the layer and toast tokens, and the motion	FR-KIOSK-07, FR-BOARD-03, NFR-USE-01, NFR-USE-04	Phase

Job	Builds	Requirements	Review
E2E0	`apps/e2e` with `@playwright/test`: Chrome through the installed browser and WebKit downloaded once; Chrome desktop and phone, WebKit desktop and iPhone, and an iPad for the kiosk; a seeded stall-mode server per browser; the smoke flow	NFR-COMP-01, NFR-MAINT-02, NFR-MAINT-04	Phase

Done when: the four gates pass; `npm run e2e` passes the smoke flow on Chrome and WebKit; a 0000 database upgrades to 0002; and the router test shows every stub route with an h1.

8.2 Phase 1: backend slices (about 3 h, still during the picks)

The API jobs build contracts, domain, db and API together, with their tests and the legacy tests they port.

Job	Builds	Requirements	Review
API-KIOSK	The gym's name and country; check-in by phone or member id, with the hour rule and the same-gym check; the name search (from the start, folded, at most 6, archived left out); join, with the coach in turn; events; the front-desk answer for an archived member, recording nothing	FR-KIOSK-01 to FR-KIOSK-06, FR-LIVE-01	Fresh
API-MEMBERS	The list by view; the detail; a PATCH of coach, phone and name that changes only the fields sent; add, by owners and coaches; archive and restore, a coach only their own; permanent delete, by owners, archived only, cascading; events	FR-MEM-01, FR-MEM-05, FR-MEMBER-01, FR-MEMBER-02, FR-MEMBER-05 to FR-MEMBER-07, FR-STAFF-07, FR-PRIV-03, FR-LIVE-02	Fresh
API-NUDGES	Templates, with create, edit, delete and render; recording a nudge with its channel and the same-gym checks; `smsUrl` and `whatsappUrl`; outcomes; the 30-day history	FR-TPL-01 to FR-TPL-03, FR-NUDGE-02, FR-NUDGE-06 to FR-NUDGE-08	Fresh
API-REPORTS	Reports, counting archived members' history and nudges by the gym's day (the Auckland case); Billing, with the pilot day from 1 to 90 and no price	FR-REPORT-01, FR-REPORT-03, FR-BILL-01	Phase
API-GYM	Sign-up's country, required and one of the five; settings GET and PATCH; the seed's country from its zone, with `TENURE_COUNTRY` and the US fallback; the kiosk and phone links from `ctx.lan`; the QR codes	FR-AUTH-01, FR-AUTH-02, FR-GYM-01 to FR-GYM-03, FR-GYM-05, FR-GLOBAL-05	Fresh
API-STAFF	Staff with their counts; deactivate and reactivate; invites for a coach or an owner, by email and a link, for 7 days; looking up and accepting one, an owner invite making an owner; the permissions sweep over every route	FR-STAFF-01 to FR-STAFF-07, FR-AUTH-06, NFR-SEC-04	Fresh
API-IMPORT	CSV parsing, the mapping and the full-name split; the date order; the planner, with phones read in the gym's country; planning and writing in one transaction, with each coach checked against the gym; restoring a matched archived member; preview and commit	FR-IMPORT-01 to FR-IMPORT-10, NFR-SEC-05	Fresh
OPS1	`npm run backup` (`VACUUM INTO` a dated file) and `npm run restore`, refused while the server runs or on a failed integrity check; `scripts/backup.sh` for the Docker volume. `npm run reset-link` moved to A1, since it needs phase 3's `password_resets` table	FR-OPS-02, NFR-REL-03	Fresh

Job	Builds	Requirements	Review
W0	Shared web parts: the member filter and search, the board's week cells in `shared/board`, the URL filter bar with chips, `useGymFormat`, and Today's clock in the gym's locale	FR-MEM-02, FR-BOARD-01, FR-GLOBAL-03	Phase

Done when: the gates and the smoke e2e flow pass; every ported test carries its ID; and the permissions sweep passes on every route.

8.3 Phase 2: core screens (about 3 h, after Uzair's design polish picks)

Job	Builds	Requirements	Review
C2	The copy ids found missing, and the NudgeButton stub	NFR-COPY-01	Phase
UI2	The design polish picks and harmony fixes, in `ui` only	None	Phase
W-SHELL	The tab bar, the More sheet and the header below 1024 px, and the sidebar from 1024 px; the owner's pilot day; Me	FR-SHELL-01 to FR-SHELL-04, FR-ME-01	Phase
W-KIOSK	The keypad, and the number at `--text-kiosk`; the name search; join; the welcome, with its pop and the plain line from day 90; the front-desk line; the idle reset; the N1 blend; the "Sample data" tag	FR-KIOSK-01 to FR-KIOSK-08	Fresh
W-TODAY-BOARD	Today's Open buttons and the line for a member with no coach; `/app/board` with chips, search, the owner's coach filter, the legend, pace and the check-in pop; "Nudge →" on the first slipping row	FR-TODAY-01, FR-TODAY-02, FR-BOARD-01 to FR-BOARD-03, FR-MEM-02	Phase
W-MEMBERS	The list in three views, a table from 48 rem and cards below; the member page with its facts, DaysGrid, check-ins, nudges with channel words, the owner's edits and NudgeButton	FR-MEM-01 to FR-MEM-04, FR-MEM-06, FR-MEMBER-01 to FR-MEMBER-04	Phase
W-CONTACT	The composer and the "Contact {first}" menu: Text message on phones only, WhatsApp and Copy message, each a link that records on the same click; the no-phone variant, with "Record nudge"	FR-NUDGE-01 to FR-NUDGE-05	Phase
W-REPORTS	The cohorts, the chart with its end labels and the nudge figures; Billing, with the pilot day, one bar and no price	FR-REPORT-01 to FR-REPORT-03, FR-BILL-01	Phase
W-NUDGES-TEMPLATES	The nudges log, with plain names, Open and channel words; templates, which owners edit with a preview and coaches read	FR-NUDGE-06, FR-TPL-01	Phase

Done when: the gates pass; `npm run e2e` passes the kiosk, live, today-board, members, contact, reports and templates flows on the desktop projects, with the live flow showing a kiosk check-in on an owner's Today in under 2 s; the kiosk, shell and contact flows also pass on the phone projects; and the speed checks pass.

8.4 Phase 3: management (about 2.5 h), then pause 1

Job	Builds	Requirements	Review
C3	The remaining copy, and the StaffPanel stub	NFR-COPY-01	Phase
W-SETTINGS	The gym form, with a Country of the five and the zone list rule; the phone and kiosk panels, with their QR codes and Copy; the "Import members or check-ins" box	FR-GYM-01 to FR-GYM-04	Phase
W-STAFF	The staff panel; the "Invite staff" dialog, with Coach or Owner, an email and the link; deactivate and "Reactivate"	FR-STAFF-02, FR-STAFF-03, FR-STAFF-05	Phase
W-AUTH	The invite page, with its owner variant; sign-up's required Country choice and its new line	FR-AUTH-01 to FR-AUTH-03, FR-STAFF-04	Phase
W-IMPORT	`/app/settings/import`: members or check-ins, as a segmented choice; a file or pasted text; the preview and column map; the date-order line and its switch; the rows as read; the problems, restored members included; the result	FR-IMPORT-01 to FR-IMPORT-05, FR-IMPORT-09, FR-IMPORT-10	Fresh
W-MEMBER-MANAGE	Add member; "Archive", "Restore member" and "Delete permanently", each with its confirm; the empty gym's links on Today	FR-MEM-05, FR-MEMBER-05 to FR-MEMBER-07, FR-TODAY-03	Fresh
A0	Migration 0003 (`password_resets` and `outbox`) and their repositories; the email stand-in writes to the outbox	FR-DEV-02, FR-OPS-04	Fresh

Done when: the gates pass, and every e2e flow passes on all projects; a 2,000-row import's check step shows in under 5 s; and the orchestrator's browser check passes, in both themes at desktop and phone width, in the Chrome pane and in iPhone Safari on the Simulator.

8.5 Phase 4: account basics and the website (about 2 h)

Job	Builds	Requirements	Review
C4	The reset words, and the home page words from the copy round	FR-AUTH-07, FR-WEB-01	Phase
A1	Password reset: the same answer whether or not the email exists; rate limited; each link working once and expiring; a reset ending the other sessions; `/reset`, and "Forgot your password?" on Login; and `npm run reset-link -- <email>`, moved here from OPS1, which prints a reset link on the stall laptop and in Docker	FR-AUTH-07, FR-AUTH-08, NFR-SEC-03	Fresh
A2	Invites sent by email through the stand-in, with the link still shown to copy	FR-STAFF-03	Fresh
A3	The `/dev/outbox` page, in development only: `TENURE_OUTBOX` is on in `npm run dev`, and off in stall mode and Docker	FR-DEV-01, NFR-SEC-06	Fresh
WEB	Home, from the copy round, and Pricing, the free pilot with no price; sign-up from the website; "Try the demo" in stall mode; the demo reset command and its stall-mode button	FR-WEB-01 to FR-WEB-05, FR-OPS-03	Fresh
OPS2	`docs/DEMO.md`, the stall runbook, which says deleted members stay in older backups; the Docker steps; a backup and restore round trip on the `tenure-data` volume	FR-OPS-02, FR-OPS-05, NFR-PRIV-02	Phase

Done when: the gates and e2e pass on all projects, with the account and website flows added; and `scripts/stall.sh`, a backup and a restore round trip work against the Docker volume.

8.6 Phase 5: close (about 2 h), then pause 2

Job	Builds	Requirements
FIN1	The trace test, flake fixes, and the full e2e run on all projects twice in a row	NFR-MAINT-03
FIN2	One fresh review of the whole version, read with a security eye, and its fixes	NFR-SEC-01 to NFR-SEC-06
FIN3	PROGRESS.md, this spec's revision history, and ARCHITECTURE §13	None

Done when (from FIN1 to FIN3 and the plan's checks): the gates, `npm run e2e` with its speed checks and the trace test pass, and the full e2e run passes on all projects twice in a row; FIN2's fixes are in; every Must has a test or a stated check; and `docs/SPEC.pdf` renders.

8.7 Side tracks

Three tracks run alongside phases 0 and 1. Each comes to Uzair as questions, in rounds of four.

Track	What happens	What it feeds
Core-flow edge cases	The five finders that didn't finish run again, limited to core flows, and the three finished ones are kept. A synthesizer writes `edge-cases.md`. Rare cases go to a later-polish list	Phase 1's jobs, before they merge
Design polish	The six reviewers run again, after the screenshots are retaken if the old ones are gone. The picks page "Tenure design polish" is drawn and checked, and Uzair picks	UI2 and the screen jobs. The picks come before phase 2
Home page copy	A copywriter offers 2 or 3 options for the headline, the line under it and the buttons, in the voice of `marketing/brand/DESIGN.md`, and Uzair picks	C4 and WEB. The pick comes before phase 4

8.8 Pauses

Pause	When	What happens
Pause 1	After phase 3, at about 10.5 h	A phone notification, and Uzair reviews the working core app
Pause 2	After phase 5, at the end of the first version	A phone notification, and Uzair reviews the first version

At each pause the orchestrator checks by hand:

- the core flows, clicked through in the Chrome pane, in both themes, at desktop and phone width;
- iPhone Safari on the Simulator, against the stall server (NFR-COMP-01);
- Docker, through `scripts/stall.sh`;
- a backup and restore round trip (FR-OPS-02).

9. Risks

Impact is rated high, medium or low for the first version. A mitigation marked "Proposed" isn't in the approved plan yet.

ID	Risk	Impact	Mitigation
R-01	Playwright may have no WebKit build for this macOS, and WebKit only stands in for Safari	Medium. The Safari half of NFR-COMP-01 would go unchecked by the e2e gate	E2E0 checked first: `npm run e2e:install` downloads WebKit once into a gitignored cache. iPhone Safari on the Simulator is checked by hand at each pause, and is the fallback if a build goes missing (NFR-COMP-01)
R-02	A design polish pick changes a ui part's layout, or the picks come late	Medium. Part of UI1 is redone, and phase 2 waits for the picks	Phase 1 is backend only, so it runs while the picks are made. The picks come before any screen is built, and the harmony fixes go into UI2 and the screen jobs
R-03	A typed stub doesn't match the two jobs that meet at it: NudgeButton's props between W-CONTACT and W-MEMBERS in phase 2, or the StaffPanel in phase 3	Medium. The member page's Nudge button, or the staff panel, breaks at the merge	The phase's copy job (C2, C3) writes the stub first, typed, so both jobs build against one shape and the typecheck fails on a mismatch
R-04	A deleted member stays in older backups	Low. A data-protection deletion isn't complete while those backups exist, and restoring one brings the member back	DEMO.md says so (NFR-PRIV-02). Backups are the operator's to keep or delete (§6.3)
R-05	Answers come late: the core-flow edge cases, the home page words, or a product question a builder hits	Medium. Phase 1's merges wait for the edge cases, and WEB for the home page words	The side tracks run alongside phases 0 and 1, in rounds of four questions (§8.7). A product question stops for Uzair, and the other jobs carry on
R-06	Six parallel jobs collide in shared files	High. Merges conflict, and one job's change breaks another's	Registration is written once, in PL0, so later jobs edit only their own folders. Each shared file has one owner per phase, and the boundary lint keeps modules and features out of each other's files (NFR-MAINT-01)
R-07	`npm run reset-link` needs the `password_resets` table, which A0 adds in phase 3, and password reset, which A1 builds in phase 4	Low. Built any earlier, the command couldn't print a working link	The command moved from OPS1 in phase 1 to A1 in phase 4, after A0's migration (FR-AUTH-08). The plan was changed on 7 Nov
R-08	The build runs long, or the usage limit stops it mid-phase	Medium. The first version comes later than "as soon as possible"	The most important flows come first, and the Shoulds only if time allows. Rare edge cases and the rest of the old tests wait for the polish. The 92% wind-down rule and a 20-minute usage check hold, and RESUME.md and PROGRESS.md stay current, so work resumes where it stopped
R-09	E2E flows are flaky across the five browser projects	Medium. A flaky gate blocks merges, or hides a real failure among the flakes	Playwright runs without retries, so a flake shows. FIN1 fixes flakes and runs the full e2e on every project twice in a row

ID	Risk	Impact	Mitigation
R-10	A security mistake in sign-in, permissions, a data write, the import, the kiosk, deletion or email	High. One gym's data reaches another, or a coach acts beyond their own members	A fresh reviewer for each of the 17 jobs that touch these. The permissions sweep checks every route against its declaration (NFR-SEC-04). FIN2 reviews the whole version with a security eye
R-11	The phone rule misreads a number: a real foreign number whose digits also make a local one is stored as local, and the phone data may not know newly assigned numbers	Low. A few members' phones are stored wrong, or refused	Accepted by Uzair, since it rescues mistyped "+" numbers (answers.md Q34). A foreign number needs only a possible length, so stale data affects only the gym's own country (§7.2.4). Updating `libphonenumber-js` is a lockfile change, not a new package (NFR-MAINT-04)
R-12	The demo gym keeps its +1 sample numbers in every country, so at a UK, Pakistani or UAE demo gym a sample number typed without "+1" is read in that country, where it isn't the member's	Low. At the stall, a demo member isn't found by phone	Typed with "+1", the numbers stay international (§7.2.4), and the name search finds them too. Proposed: DEMO.md says so (FR-OPS-05)
R-13	Behind Docker every device counts as one for the rate limits (ARCHITECTURE §12)	Medium. At a busy stall the kiosk's 30 requests a minute, and the 10 failed logins per account, are shared by everyone	Proposed: one kiosk per Docker stall, or `npm run stall`, which sees each device's own address. DEMO.md says which
R-14	The stall demo ages: its dates are relative to the day it was seeded, and each visitor sees what earlier visitors entered	Medium. An old demo shows its members' weeks and statuses moved on, and a visitor sees another's name and phone	Re-seed on the morning of the stall (ARCHITECTURE §12), and between visitors with `npm run demo:reset` or the stall-mode button (FR-OPS-03). DEMO.md has the steps (FR-OPS-05)
R-15	The hand-off to Messages and WhatsApp can't be automated: Playwright can't follow a link into another app, and the Simulator has no WhatsApp	Medium. A hand-off that fails on a real phone, the case FR-NUDGE-03 guards against, would go unseen	Each menu item is a link that records on the same click (FR-NUDGE-03), and the flows check each link and its record. Proposed: open each item on a real iPhone and an Android phone at pause 1
R-16	The kiosk link is the gym's only key to the kiosk	Medium. Anyone with the link can see members' first names and initials, and check members in	The search shows at most 6 names, as "First L.", and leaves archived members out, and the kiosk takes 30 requests a minute (FR-KIOSK-02, NFR-SEC-03). Changing the token isn't in the first version, since it would break printed links

10. Acceptance tests

The first version is accepted when every Must in §3 passes these tests on Chrome and on Safari, through Playwright's WebKit (NFR-COMP-01), the §4 checks below pass, and the checks in §10.15 have been made by a person. Each automated test's name starts with the ID it covers (NFR-MAINT-03), so one row here can be several tests.

How the tests run:

- `npm run e2e` runs the 13 Playwright flows, one file each in `apps/e2e/src`. The projects are Chrome desktop and phone, through the installed Chrome, and WebKit desktop and iPhone, standing in for Safari. The kiosk flow runs on an iPad and on the two phone projects, so it runs in both engines too.
- Each engine has its own stall-mode server, with a freshly seeded demo gym in America/Chicago, so the gym is in the US (FR-GLOBAL-05). Demo logins come from the seed module, never typed into a test. A test-only factor lifts the rate limits, so the limits themselves are checked by API tests.
- Flows run in parallel against the same server. A flow that changes data uses a gym it signs up, or members and staff it adds, so flows don't meet. A step done "through the API" calls the API directly, as the signed-in user.
- A speed check times from the navigation until the screen's data shows, not just its title.
- API tests run in `npm test`, against an in-memory database with a fixed clock. Their paths are under `apps/api/src`; other paths are given in full.
- In "Expected result", "1:" answers step 1.

10.1 Smoke (`smoke.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-01	The owner logs in	1. Open `/login`. 2. Enter the demo owner's email and password, and press "Log in".	Today opens at `/app`, titled "Today".	`smoke.e2e.ts`	FR-AUTH-04
TC-02	A coach logs in	1. Log in as the demo coach, Priya S. 2. Read the "Slipping now" rows.	Today is titled "My members". Every row names Priya as the coach, and no other staff member.	`smoke.e2e.ts`	FR-TODAY-01
TC-03	Today loads in time	1. Log in as the owner. 2. Open `/app`, timing until the KPIs and "The board" show.	Under 2 s.	`smoke.e2e.ts`	NFR-PERF-01

10.2 Kiosk (`kiosk.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-04	The kiosk opens without signing in	1. In a signed-out browser, open the demo gym's kiosk link. 2. Open `/kiosk/` with a wrong token.	1: "Check in", the gym's name and the "Sample data" tag. 2: a not-found screen.	`kiosk.e2e.ts`; `modules/kiosk/kiosk.test.ts`	FR-KIOSK-01, FR-KIOSK-07

ID	Test	Steps	Expected result	Covered by	Covers
TC-05	Check in by phone	1. Type a seeded member's number on the keypad, without "+1", and press "Check in". 2. Type it again with "+1" first. 3. Check in a seeded member who joined over 90 days ago. 4. Type a number no member has.	1: the number formats as it's typed. "Welcome back, {first}. Week {n}." shows with the visits line, under 1 s after the press. 2: the same member. 3: "Welcome back, {first}.", with no week, and the visits line. 4: "We couldn't find that number. Try your name, or join as a new member." Nothing is stored.	`kiosk.e2e.ts`; `modules/kiosk/kiosk.test.ts`	FR-KIOSK-02, FR-KIOSK-05, FR-GLOBAL-02, NFR-PERF-02
TC-06	A visitor joins	1. Press "New here? Join", then "Join and check in" with no first name. 2. Join as "Zoë", with a new phone and no last name. 3. At once, check in with that phone. 4. Join again with a phone a member already has.	1: "Enter your first name." 2: "Welcome, Zoë. Day 1 of 90." Zoë is a member, joined today and checked in, with a coach. 3: welcomed, and Zoë still has one check-in. 4: refused, and nothing is stored. In the API test, each join gets the next active staff member as coach, in turn.	`kiosk.e2e.ts`; `modules/kiosk/kiosk.test.ts`	FR-KIOSK-03, FR-KIOSK-04, FR-KIOSK-05, FR-GLOBAL-04
TC-07	Check in by name	1. Press "Use my name" and type "ZOE", then "oe". 2. Type "a". 3. Pick a name.	1: "ZOE" finds "Zoë", shown alone. "oe" finds nobody, since the search matches from the start. 2: at most 6 people, each shown as "First L.". 3: that member's welcome.	`kiosk.e2e.ts`; `modules/kiosk/kiosk.test.ts`	FR-KIOSK-02, FR-GLOBAL-04
TC-08	An archived member at the kiosk	1. Join with a new first name and phone, then archive that member as the owner, through the API. 2. Search their first name. 3. Type their phone and press "Check in". 4. Join again with their phone.	2: they aren't listed. 3 and 4: "Please see the front desk to check in." Nothing is recorded: no check-in and no new member.	`kiosk.e2e.ts`; `modules/kiosk/kiosk.test.ts`	FR-KIOSK-02, FR-KIOSK-06
TC-09	A hardware keyboard, and the idle reset	1. Type a seeded member's number on the keyboard and press Enter. 2. Type a few digits, then leave the kiosk alone for 30 s on Playwright's clock.	1: the welcome, as with the keypad. 2: the kiosk is back at its start, with the number cleared.	`kiosk.e2e.ts`	FR-KIOSK-08, NFR-USE-02
TC-10	The kiosk's look and sizes	1. Open the kiosk in the dark theme, then the light one. 2. Type a long number, then measure the text, the keys and their outlines.	1: the blend's pattern differs between the themes. 2: no text is over 64 px except the typed number, which is up to 128 px. Every key is at least 64 px, outlined in `line-blend`.	`kiosk.e2e.ts`; `packages/ui/src/Button/button-size.node.test.ts`	FR-KIOSK-07, NFR-USE-03
TC-11	The welcome's pop	1. Check in. 2. Emulate reduced motion, and check in again.	1: the welcome line enters with a pop, scaling from 1.25 to 1 over 320 ms. 2: nothing moves.	`kiosk.e2e.ts`	FR-KIOSK-05, NFR-USE-04

10.3 Live updates (`live.e2e.ts`)

The live flow uses two browser contexts in one engine, so both reach the same server.

ID	Test	Steps	Expected result	Covered by	Covers
TC-12	A join at the kiosk reaches Today	1. Log in as the owner, on Today. 2. In a second context, a visitor joins at the kiosk.	Within 2 s, with no reload, "Checked in today" lists the visitor, the toast "{name} checked in" shows, and "New members" counts them.	`live.e2e.ts`	FR-TODAY-04, FR-LIVE-01, NFR-PERF-03
TC-13	The board's week cell pops	1. Log in as the owner, on the board. 2. At the kiosk, check in a member the flow added. 3. Emulate reduced motion, and do it again with another.	2: the member's week cell fills with a pop, scaling from 1.25 to 1 over 320 ms. 3: the cell fills, and nothing moves.	`live.e2e.ts`; `packages/ui/src/motion/pop.node.test.ts`	FR-BOARD-03, NFR-USE-04
TC-14	Each coach hears their own members	1. Log in as a coach, on Today. 2. At the kiosk, check in a member the flow added for another coach, then one added for this coach.	Only this coach's member shows, with its toast. In the API test, owners get every event, and a member moved from one coach to another reaches both.	`live.e2e.ts`; `modules/realtime/scope.test.ts`	FR-LIVE-02, FR-STAFF-07
TC-15	An import refreshes Today once	1. Sign up a gym, and import a file of 12 members. 2. Open Today in a second context, and count its data requests. 3. In the first, import a file of their 135 check-ins.	Today refreshes once for the import. In the web test, a stream that drops and reopens refreshes each screen once too.	`live.e2e.ts`; `apps/web/src/shared/live/live.test.ts`	FR-LIVE-03, FR-IMPORT-10
TC-16	Every write publishes its event	Through the API, make each write: a check-in, a join, a member edited, archived, restored and deleted, a nudge, an import, a staff change and a settings change.	Each publishes its event once it succeeds, and an owner's stream receives it. Gym-wide events reach everyone in the gym, and nobody else.	`modules/realtime/realtime.test.ts`; `modules/realtime/scope.test.ts`	FR-LIVE-01, FR-NUDGE-08
TC-17	The stream ends with the session	Through the API, open a stream, then: 1. log out; 2. deactivate the user; 3. stop the server.	1 and 2: the stream gets no more events, and closes. 3: it closes, and the server exits within about a second; a server that hangs is stopped by the 3 s fallback.	`modules/realtime/realtime.test.ts`; `stall.test.ts`; `platform/shutdown.test.ts`	FR-LIVE-04, FR-AUTH-06, NFR-REL-01

10.4 Today and the board (`today-board.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-18	Today's parts	1. Log in as the owner, on Today.	The four KPIs ("New members", "On the line", "Slipping" and "Nudged this week"), "Slipping now", "Checked in today" with times in the gym's US locale, such as "7:42 AM", and "The board". In the API test, the read model agrees with the seed's own oracle.	`today-board.e2e.ts`; `modules/today/agreement.test.ts`	FR-TODAY-01, FR-GLOBAL-03

ID	Test	Steps	Expected result	Covered by	Covers
TC-19	"Slipping now"	1. As the owner, who also coaches, read "Slipping now". 2. Add a member who joined three weeks ago, and set their coach to "No coach".	1: the owner's own members come first, then the most days since the last visit. Each row has "Nudge →". 2: the new member is slipping, and their row shows "No visits yet" alone.	`today-board.e2e.ts`; `modules/today/today.test.ts`	FR-TODAY-02
TC-20	The board	1. Open the board, timing until its rows show. 2. Go back to Today, and read its board.	1: every member in their first 90 days, slipping first, then the newest joiners. Each row has 13 week marks, the pace, the status and a quiet "Open" named for the member, such as "Open Sam O.". The legend shows. Under 2 s. 2: the same order, and its first slipping row also has "Nudge →".	`today-board.e2e.ts`; `packages/domain/src/board.test.ts`	FR-BOARD-01, FR-BOARD-02, FR-MEM-03, NFR-PERF-01
TC-21	The filter bar	1. On the board, search "SAM", then a seeded member's phone typed with a leading 0. 2. Pick the "Slipping" chip. 3. As the owner, pick a coach. 4. Reload, then open Members.	1: case is folded, and the leading 0 is ignored. 2: only slipping members. Each chip has its status dot. 3: the coach filter offers only the coaches of the members on screen, and narrows to theirs. 4: the URL keeps every filter, and Members has the same bar. In the domain test, accents fold too.	`today-board.e2e.ts`; `packages/domain/src/text.test.ts`	FR-MEM-02, NFR-USE-05
TC-22	A coach's board	1. Log in as a coach, and open the board.	Only their own members, with no coach filter and no coach column.	`today-board.e2e.ts`	FR-STAFF-06, FR-STAFF-07
TC-23	The board with a keyboard	1. With Tab, Enter and Space only, pick a chip, type a search and open a member.	Each step works, and focus always shows.	`today-board.e2e.ts`	NFR-USE-02

10.5 Members and the member page (`members.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-24	The three views	1. Open Members, timing until the list shows. 2. Choose "Everyone", then "Archived".	1: "First 90 days" is chosen, and lists the members in their first 90 days, in under 2 s. 2: "Everyone" adds the members past day 90 and leaves out archived ones. "Archived" lists only archived members.	`members.e2e.ts`; `modules/members/members.test.ts`	FR-MEM-01, NFR-PERF-01
TC-25	The list and its names	1. Add a member with the first name "Leo" and no last name. 2. Read Members on a desktop project, then on a phone project.	A table from 48 rem, and cards below. Names are plain text, slipping ones in the attention colour, and each row ends with a quiet "Open" named for the member. Names show as "First L.", and the new member as "Leo" alone, before any "Leo" with a last name. The order follows the gym's locale.	`members.e2e.ts`	FR-MEM-03, FR-MEM-04, FR-GLOBAL-04

ID	Test	Steps	Expected result	Covered by	Covers
TC-26	Add a member	1. As the owner, press "Add member", and fill in "Add a member": a first name, a phone, a joined date and a coach. Press "Add member". 2. Try again with a phone another member has, then with a joined date after today. 3. As a coach, add a member.	1: "{name} added", and the member is listed. 2: both are refused. 3: the new member is the coach's.	`members.e2e.ts`; `modules/members/members.test.ts`	FR-MEM-05
TC-27	The empty states	1. Search for a name nobody has. 2. Open Members in a gym signed up for the test.	1: "No members found", with "Try another name or clear the filters." 2: "Members are listed here once they join at the front desk or are imported."	`members.e2e.ts`	FR-MEM-06
TC-28	The member page	1. Press "Open" on a seeded member who has nudges, timing until the page shows. 2. Open `/app/members/` with an id nobody has.	1: the name, coach, phone and joined date, "The 90 days" grid, the check-ins, and the nudges, each with its channel word ("Text", "WhatsApp", "Copied" or "In person"). Under 2 s. 2: "Member not found".	`members.e2e.ts`	FR-MEMBER-01, NFR-PERF-01
TC-29	The phone's format	1. At the demo gym, open a seeded member, whose number is +1. 2. At a UK gym signed up for the test, add a member with a UK mobile, and one with a "+1" number.	1: national, such as "(312) 555-0142". 2: the UK mobile national, such as "07911 123456", and the +1 number international, such as "+1 312 555 0142".	`members.e2e.ts`; `packages/domain/src/phone-format.test.ts`	FR-MEMBER-03, FR-GLOBAL-02
TC-30	The owner's edits	1. As the owner, change the name, coach (to "No coach") and phone of a member the flow added, each with its own "Change". 2. Clear the first name, then enter a phone another member has.	1: each saves, and each request sends only its own field. 2: each is refused, with the server's error under the field: "Enter the member's first name." and "That email or phone number is already in use. Check it and try again."	`members.e2e.ts`; `modules/members/members.test.ts`	FR-MEMBER-02
TC-31	A coach's limits	1. As a coach, open one of their members. 2. Open another coach's member by its URL.	1: no "Change" buttons, and each edit answers 403 through the API. 2: "Member not found", and the API answers 404.	`members.e2e.ts`; `modules/members/members.test.ts`	FR-STAFF-06, FR-STAFF-07

10.6 Contact and the nudges log (`contact.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-32	The composer	1. As a coach, press "Nudge →" on a "Slipping now" row. 2. Pick a template, then "Write my own", and edit the message. 3. Press "Nudge →" in a member page's head row, then on the first slipping row of Today's board.	1: "Nudge {first}" opens. 2: a template fills `{first}`, `{coach}` with the coach's first name, `{gym}` and `{days}`, and the message stays editable. 3: each opens the same composer, for that member.	`contact.e2e.ts`; `packages/domain/src/templates.test.ts`	FR-NUDGE-01, FR-TPL-02, FR-TODAY-02, FR-MEMBER-04, FR-BOARD-02
TC-33	The Contact menu on a phone	1. On a phone project, press "Contact {first}". 2. Pick "Text message".	1: "Text message", "WhatsApp" and "Copy message", each a link. 2: the link is `sms:` with the member's number and the message. The same click records the nudge: "Nudge to {first} recorded" shows, and the log says "Text".	`contact.e2e.ts`; `packages/domain/src/sms.test.ts`	FR-NUDGE-02, FR-NUDGE-03
TC-34	WhatsApp and Copy message	1. Pick "WhatsApp"; the test stops the navigation to `wa.me`. 2. Open the composer again, and pick "Copy message". 3. On a desktop project, open the menu.	1: a `wa.me` link with the number and the message, recorded as WhatsApp on the same click. 2: the clipboard holds the message, recorded as copied. 3: no "Text message".	`contact.e2e.ts`; `modules/nudges/nudges.test.ts`	FR-NUDGE-02, FR-NUDGE-03
TC-35	No phone on file	1. Open the composer for a member who joined at the kiosk without a phone. 2. Press "Record nudge".	1: "No phone on file", and no Contact menu. 2: the nudge is recorded, and the log says "In person".	`contact.e2e.ts`	FR-NUDGE-04
TC-36	Tenure sends nothing itself	1. Record every request the browser makes during the flow.	Every request goes to Tenure's own server. A message leaves only through a link to the coach's own app, or the clipboard.	`contact.e2e.ts`; `adapters/adapters.test.ts`	FR-NUDGE-05, NFR-PRIV-01
TC-37	Recording checks the gym	Through the API, record a nudge for another gym's member, then with another gym's template, then as a coach for another coach's member.	Each answers not found, and nothing is recorded. A good nudge is stored and publishes its event.	`modules/nudges/nudges.test.ts`	FR-NUDGE-08, FR-PRIV-02, FR-STAFF-07

ID	Test	Steps	Expected result	Covered by	Covers
TC-38	The nudges log	1. As the owner, open Nudges, timing until the rows show. 2. As a coach, open Nudges.	1: the last 30 days. Each row has the member's plain name and "Open", the start of the message, the date in the gym's locale, the sender, the channel word, and "Came back", "No visit yet" or "No visit in 7 days". Under 2 s. 2: "Your nudges in the last 30 days.", with only theirs. In the API test, a nudge from before the first version has no channel word. In the domain test, a check-in at the nudge's instant doesn't come back, and ones a minute and exactly 7 days after do.	`contact.e2e.ts`; `modules/nudges/nudges.test.ts`; `packages/domain/src/metrics.test.ts`	FR-NUDGE-06, FR-NUDGE-07, FR-STAFF-07, NFR-PERF-01
TC-39	Contact with a keyboard, and target sizes	1. Open the menu with Enter, move with the arrow keys, and pick with Enter. 2. Measure the composer's buttons and the menu's items.	1: it works with the keyboard alone. 2: every target is at least 44 px.	`contact.e2e.ts`	NFR-USE-02, NFR-USE-03

10.7 Reports and Billing (`reports.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-40	Reports	1. As the owner, open Reports, timing until the figures show. 2. As a coach, read the navigation, and ask for Reports' and Billing's data through the API.	1: "Cohorts", with the share still coming at day 30, 60 and 90 for each join month, and months in the gym's locale; "Visits by week"; and "Nudge results", with the sent, came back and rate figures. Under 2 s. 2: neither page is in the navigation, and both answer 403. In the API test, the figures equal the domain's report over the seed.	`reports.e2e.ts`; `modules/reports/reports.test.ts`	FR-REPORT-01, FR-REPORT-03, FR-STAFF-06, NFR-PERF-01
TC-41	The chart	1. Read the visits-by-week chart.	Still coming is a solid line in the text colour. Slipped is dashed, in the slipping colour, with point markers. The 3-a-week line is dotted, in soft text. Each line is labelled at its end ("Still coming", "Slipped" and "The line · 3 a week"), with no legend, and the chart has a description for screen readers.	`reports.e2e.ts`	FR-REPORT-02, NFR-USE-05
TC-42	What Reports count	In the API test: 1. archive a member with history, then delete them; 2. at a gym far east of UTC (the Auckland case), send a nudge late on the gym's day.	1: archived, their history still counts in every figure; deleted, it doesn't. 2: the nudge counts by the gym's day, not by UTC's.	`modules/reports/reports.test.ts`	FR-REPORT-01, FR-REPORT-03, FR-MEMBER-07

ID	Test	Steps	Expected result	Covered by	Covers
TC-43	Billing	1. As the owner, open Billing, timing until it shows.	"Pilot day {n} of 90", counted from the gym's pilot start, the 90 days as one bar, and "Free 90-day pilot.". No price and no plan button. Under 2 s. In the API test, the day is 1 on the start day and never over 90.	`reports.e2e.ts`; `modules/billing/billing.test.ts`	FR-BILL-01, NFR-PERF-01

10.8 Templates (`templates.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-44	The owner edits templates	1. As the owner, open Templates, timing until the list shows. 2. Press "New template", and type a name and a message with `{first}`, `{coach}`, `{gym}` and `{days}`. 3. Press "Save template", edit it, and save again. 4. Press "Delete", then "Delete template".	1: the gym's templates, in under 2 s. 2: the preview fills the four variables for the sample member as it's typed. 3: "Template saved" each time. 4: "Delete this template" asks first; then "Template deleted", and it's gone.	`templates.e2e.ts`	FR-TPL-01, FR-TPL-02, NFR-PERF-01
TC-45	Coaches read, and the limits	1. As a coach, open Templates. 2. As a coach, create, edit and delete a template through the API. 3. As the owner, through the API, save a name of 61 characters, then a message of 1,001.	1: the templates, and "Your owner edits templates. You can use them when you nudge.", with no editor. 2: each answers 403. 3: both refused; 60 and 1,000 characters are saved.	`templates.e2e.ts`; `modules/templates/templates.test.ts`	FR-TPL-01, FR-STAFF-06

10.9 The app shell and Me (`shell.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-46	Below 1024 px	1. On a phone project, log in as the owner. 2. Open More, then a page in its sheet. 3. Measure the tabs, the sheet's items and the header's buttons.	1: the header shows the gym name, the theme toggle and "Log out", with no logo. The tab bar holds Today, Board, Members, Nudges and More. 2: the sheet holds the owner's other pages, and More stays marked while that page is open. 3: each is at least 44 px.	`shell.e2e.ts`	FR-SHELL-01, NFR-USE-03
TC-47	From 1024 px	1. On a desktop project, log in as the owner. 2. Measure the sidebar's links and buttons.	1: a sidebar holds the logo, the gym name and every page, with the theme toggle and "Log out" at its foot. 2: each is at least 44 px.	`shell.e2e.ts`; `packages/ui/src/Button/button-size.node.test.ts`	FR-SHELL-02, NFR-USE-03
TC-48	The pilot day by the gym name	1. As the owner, look at 1280 px, then 800 px, then on a phone. 2. As a coach, at 1280 px.	1: "Pilot day {n} of 90" under the gym name in the sidebar, then after the gym name in the header, and not on the phone. 2: not shown.	`shell.e2e.ts`	FR-SHELL-03

ID	Test	Steps	Expected result	Covered by	Covers
TC-49	Owner pages, and signed-out visitors	1. Log in as a coach, and read the navigation. 2. Signed out, open `/app` and `/app/board`.	1: no Reports, Settings or Billing. 2: each goes to `/login`.	`shell.e2e.ts`	FR-SHELL-04, FR-STAFF-06
TC-50	"Try again"	1. Make Today's data request fail once; the test routes it to an error. 2. Press "Try again".	1: the screen offers "Try again". 2: Today's data loads.	`shell.e2e.ts`	FR-SHELL-05
TC-51	Focus after a navigation	1. With the keyboard, follow a navigation link.	Focus moves to the new page's title, marked by a short focus bar beside it instead of the ring.	`shell.e2e.ts`; `apps/web/src/shared/page/page.test.tsx`	FR-SHELL-06, NFR-USE-02
TC-52	The themes	1. Emulate a dark device, then a light one. 2. Switch with the toggle, and reload. 3. Read the theme at the reload's first paint. 4. Emulate reduced motion, and switch.	1: Navy & Lime, then Summer Ocean Breeze, by the page's `bg`. 2: the toggle's choice stays on that device. 3: the chosen theme is set before the app's code runs. 4: the switch is instant.	`shell.e2e.ts`; `apps/web/src/vite-plugins/theme-boot.test.ts`	FR-SHELL-07, NFR-USE-04
TC-53	Me, and log-out	1. Open Me. 2. Press "Log out", then open `/app` again.	1: the user's name, email, gym and role, the "Theme" choice (Light, Dark and Same as device), and "Log out". 2: `/login`, and the old session is refused.	`shell.e2e.ts`; `modules/auth/auth.test.ts`	FR-ME-01, FR-AUTH-05
TC-54	No request leaves the machine	1. Record every request while opening the website, each app page and the kiosk.	Every request goes to Tenure's own server: no outside fonts, analytics or trackers.	`shell.e2e.ts`	FR-PRIV-01, NFR-PRIV-01
TC-55	English only	1. Open each page. 2. Run the copy tests.	1: each page is marked `lang="en"`. 2: every string comes from COPY.md, which is English only, in sentence case, and no line quotes a button.	`shell.e2e.ts`; `packages/copy/src/index.test.ts`	FR-GLOBAL-06, NFR-COPY-01

10.10 Settings and staff (`settings-staff.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-56	Edit the gym	1. Sign up a gym for the test, and open Settings, timing until it shows. 2. Change the name, set the country to the UK and the zone to Europe/London, and save. 3. Have a visitor join at the gym's kiosk, then open Today.	1: under 2 s. 2: "Settings saved", and the new name shows in the shell at once. 3: "Checked in today" shows the time on the UK's 24-hour clock.	`settings-staff.e2e.ts`; `modules/settings/settings.test.ts`	FR-GYM-01, FR-GLOBAL-03, NFR-PERF-01

ID	Test	Steps	Expected result	Covered by	Covers
TC-57	The Country and zone lists	1. Sign up a US gym from a browser in a US zone that isn't one of the old app's 7, and open Settings' choices. 2. Choose Canada, and open the zones. 3. Through the API, save a country outside the five, then a zone not on the list.	1: the five markets; and the old app's 7 zones, plus the gym's own. 2: every Canadian zone, by city, east to west. 3: "Choose a country from the list." and "Choose a timezone from the list."	`settings-staff.e2e.ts`; `packages/domain/src/zones.test.ts`	FR-GYM-01, FR-GYM-02, FR-GLOBAL-01
TC-58	The links, and the import box	1. As the owner, in the Kiosk section, press "Copy link", then "Open the kiosk". 2. Read "Open on a phone". 3. Press "Import members or check-ins".	1: the kiosk link with its QR code, "Kiosk link copied", and the kiosk opens. 2: the e2e server listens on 127.0.0.1 only, so it says "No network address found. Connect this computer to Wi-Fi, then reload the page." 3: `/app/settings/import`. In the API test, with a LAN address, the kiosk and phone links both use it, each with a QR code.	`settings-staff.e2e.ts`; `modules/settings/settings.test.ts`	FR-GYM-03, FR-GYM-04
TC-59	The staff list	1. As the demo owner, read Settings' Staff section.	Each person with their role, "{n} members" in the 90-day window, and "{n} slipping" in the attention colour. In the API test, the counts are right, the owner included.	`settings-staff.e2e.ts`; `modules/staff/staff.test.ts`	FR-STAFF-02
TC-60	Invite a coach	1. In a gym signed up for the test, press "Invite staff", enter a name and an email, choose Coach, and press "Create link". 2. Press "Copy link". 3. In a second context, open the link, enter a password, and press "Join {gymName}". 4. Open the link again.	1: "We emailed this link to {email}. It works once and expires on {date}." 2: "Invite link copied". 3: the new coach is signed in, on "My members". 4: "This link doesn't work". In the API test, the email is in the outbox, the link stops working after 7 days, and in stall mode it uses the LAN address.	`settings-staff.e2e.ts`; `modules/invites/invites.test.ts`	FR-STAFF-01, FR-STAFF-03, FR-STAFF-04, FR-DEV-02
TC-61	Invite an owner, over an old session	1. Invite with Owner chosen. 2. In a context signed in as another user, open the link and accept it. 3. Send a request with that user's old cookie.	1: the invite page says "Your owner invited you as a co-owner." 2: the new person is an owner, with the owner pages. 3: refused, since accepting ended the old session.	`settings-staff.e2e.ts`; `modules/invites/invites.test.ts`	FR-STAFF-01, FR-STAFF-03, FR-STAFF-04

ID	Test	Steps	Expected result	Covered by	Covers
TC-62	Deactivate and reactivate	1. With the invited coach signed in in a second context, the owner presses "Deactivate" on their row, and confirms in "Deactivate {name}". 2. The coach goes on working, then logs in again. 3. The owner presses "Reactivate", and the coach logs in. 4. Through the API, deactivate the gym's only active owner.	1: the row shows "Inactive". 2: their session has ended, so they're sent to `/login`, and the log-in is refused. 3: they're in, with their members and history as before. 4: "The last active owner can't be deactivated."	`settings-staff.e2e.ts`; `modules/staff/staff.test.ts`	FR-STAFF-02, FR-STAFF-05, FR-AUTH-06

10.11 Import (`import.e2e.ts`)

Each step works in a gym the flow signs up.

ID	Test	Steps	Expected result	Covered by	Covers
TC-63	An empty gym leads to the import	1. Sign up a gym, and read Today. 2. Press "Import members".	1: "No members yet", with "Import members" and "Open the kiosk". 2: the import page.	`import.e2e.ts`	FR-TODAY-03
TC-64	Members, mapped and checked	1. Choose "Members" in the segmented choice, and choose a members CSV file. 2. Change one suggested column. 3. Open Members in another tab. 4. Paste members with no join date column, then check-ins with no ID or phone column.	1: a column suggested for each field from the headers, and the first rows. 2: the preview follows. 3: still empty, since nothing is saved before the commit. 4: "We couldn't find a join date column. Choose which column holds it." and "We couldn't find a member ID or phone column to match check-ins to members. Choose one."	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-01, FR-IMPORT-02, FR-IMPORT-08
TC-65	A full name column	1. Paste members under "Full name": "Maria Lopez", "Lopez, Maria" and "Leo".	The first two read as Maria Lopez, and the third as Leo, with no last name.	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-03, FR-GLOBAL-04
TC-66	The date order	1. At the US gym, paste join dates such as 03/04/2026. 2. Press "Read them day first". 3. Paste a file with a date such as 25/03/2026.	1: "Dates are read month first." and "Set from your gym's country. Check the first rows below.", with 03/04/2026 read as 4 March. 2: "Dates are read day first.", the rows read again (3 April), and the button now says "Read them month first". 3: "Dates are read day first." and "Set from the file: some of its dates only read this way."	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-04

ID	Test	Steps	Expected result	Covered by	Covers
TC-67	Problems by row	1. Paste members with a bad phone, a bad email, a coach email no staff member has, and a row whose ID matches one member while its phone is another's.	Each problem is listed as "Row {n}: ...". The bad phone and email are left blank, the unknown coach leaves no coach, and the other member's phone is left as it was. Every one of those members is kept.	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-05
TC-68	Matching, and skipped rows	1. Import a members file. 2. Import it again, with one row changed, one cell blanked and one row repeated.	2: the changed row updates only what differs, matched by ID, or by phone where it has none. The blank doesn't overwrite. Unchanged rows and the repeat are skipped ("already up to date." and "this member already appears earlier in the file.").	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-06
TC-69	The commit and its result	1. With Today open in another tab, press "Import members".	Every row is written at once. The panel shows "{created} new, {updated} updated, {skipped} skipped", the toast "Import finished" shows, and Today refreshes once. In the API test, a clash writes nothing, and a coach from another gym is never assigned.	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-08, FR-IMPORT-10, FR-LIVE-03
TC-70	Check-ins	1. Choose "Check-ins", and import check-ins for the gym's members, with one in the future, one before its member joined, and one within an hour of another.	Each matches its member by ID, then by phone. The three are skipped, each with its reason, and the rest are stored.	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-07
TC-71	An archived member comes back	1. Archive an imported member. 2. Import their row again.	2: the check step says "this member was archived, so importing restores them.", and after the commit they're back in Members.	`import.e2e.ts`; `modules/import/import.test.ts`	FR-IMPORT-09
TC-72	The file's size, and owners only	Through the API: 1. preview a CSV over 2 MB; 2. send an import request over 4 MB; 3. import as a coach.	1: "That file is over 2 MB. Choose a CSV of 2 MB or less." 2: 413. 3: 403.	`modules/import/import.test.ts`	FR-IMPORT-01, FR-STAFF-06, NFR-SEC-05
TC-73	A 2,000-row check step in time	1. Choose a generated file of 2,000 members, timing until the check step shows.	Under 5 s.	`import.e2e.ts`	NFR-PERF-04

10.12 Archive and delete (`archive-delete.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-74	Archive	1. As the owner, open a member the flow added, and press "Archive". 2. Press "Archive member" in "Archive this member?".	"{name} archived". The member leaves Today, the board, "First 90 days" and "Everyone", and is listed in "Archived".	`archive-delete.e2e.ts`	FR-MEMBER-05, FR-MEM-01

ID	Test	Steps	Expected result	Covered by	Covers
TC-75	Restore	1. Open the member from "Archived", and press "Restore member".	"{name} restored", and they're back in every list, with their history.	`archive-delete.e2e.ts`	FR-MEMBER-06
TC-76	A coach archives their own	1. As a coach, archive one of their members, then restore them. 2. Through the API, archive and restore another coach's member.	1: both work. 2: both answer not found.	`archive-delete.e2e.ts`; `modules/members/members.test.ts`	FR-MEMBER-05, FR-MEMBER-06, FR-STAFF-07
TC-77	Delete permanently	1. Archive a member the flow added, who has a check-in and a nudge. 2. Press "Delete permanently", then "Delete member" in "Delete this member for good?".	2: the confirm says their visits and nudges are deleted too. Then "{name} deleted", the screen goes back to Members, the member's page shows "Member not found", and their nudge is gone from the log.	`archive-delete.e2e.ts`	FR-MEMBER-07, FR-PRIV-03
TC-78	Delete's rules	Through the API: 1. delete an active member; 2. delete an archived one as a coach; 3. delete an archived one as the owner.	1: "Archive this member before deleting them." 2: refused. 3: the member, their check-ins and their nudges are gone from the database.	`modules/members/members.test.ts`	FR-MEMBER-07, NFR-PRIV-02

10.13 Account and website (`account-website.e2e.ts`)

ID	Test	Steps	Expected result	Covered by	Covers
TC-79	The website	1. Open `/`, then `/pricing`, recording the files they load. 2. Follow each page's way to sign up.	1: Home offers the free pilot, in the words from the home page copy round. Pricing describes the free 90-day pilot and what it includes. Neither shows a price or loads the app's code, and each shows in under 2 s. 2: `/signup`.	`account-website.e2e.ts`	FR-WEB-01, FR-WEB-02, FR-WEB-03, FR-WEB-05, NFR-PERF-01
TC-80	Sign up	1. In a browser set to Canada, in America/Toronto, open `/signup`. 2. Fill in the gym's name, a name, an email and a password of 8 characters, and press "Start the pilot".	1: "Start a free 90-day pilot for your gym.", and the Country choice lists the five markets, set to Canada. 2: an empty Today, for the new owner. Settings shows Canada and America/Toronto, and Templates the three default templates.	`account-website.e2e.ts`; `modules/auth/auth.test.ts`	FR-AUTH-01, FR-AUTH-02, FR-AUTH-03, FR-TPL-03, FR-GLOBAL-01
TC-81	Sign-up's refusals	1. In a browser set to India, open `/signup`. 2. Submit with no country. 3. Through the API, sign up with a password of 7 characters, then 201, then a country outside the five.	1: no country is chosen. 2: "Choose a country from the list." 3: each is refused, the long password with "That password is too long. Use 200 characters or fewer." and the country with "Choose a country from the list."	`account-website.e2e.ts`; `modules/auth/auth.test.ts`	FR-AUTH-01, FR-AUTH-02

ID	Test	Steps	Expected result	Covered by	Covers
TC-82	Log-in's answers	1. Log in with a wrong password, then with an email nobody has.	Both get "That email and password don't match." In the API test, the unknown email takes as long to answer as the wrong password.	`account-website.e2e.ts`; `modules/auth/auth.test.ts`	FR-AUTH-04
TC-83	The session, and passwords	1. Log in, and read the cookie. 2. In the API tests, present an expired session, serve over HTTPS, and read what's stored.	1: `tenure_session` is httpOnly and SameSite=Lax, for 30 days. 2: the expired session is refused; over HTTPS the cookie is Secure; the password is stored as scrypt, and the session token as a hash.	`account-website.e2e.ts`; `platform/auth/session.test.ts`; `platform/auth/password.test.ts`	FR-AUTH-05, NFR-SEC-01
TC-84	Password reset	1. As an owner the flow signed up, also signed in in a second context, press "Forgot your password?" on Login, enter the email, and press "Send reset link". 2. Do the same with an email nobody has. 3. Print a link with `npm run reset-link -- <email>`, open it, enter a new password, and press "Save password". 4. Open the link again.	1 and 2: the same answer, "If an account uses that email, a reset link is on its way. It works once, for an hour." 3: "Choose a new password", then "Password saved". The new password logs in and the old one doesn't, and the second context's session has ended. 4: "This link doesn't work". In the API test, a link over an hour old is refused, and only an email with an account gets a reset email in the outbox.	`account-website.e2e.ts`; `modules/reset/reset.test.ts`	FR-AUTH-07, FR-AUTH-08
TC-85	The outbox is for development	1. Open `/dev/outbox` on the stall-mode server. 2. In the API test, turn the outbox on, as `npm run dev` does, then off.	1: not found. 2: on, it lists the stand-in emails; off, its route answers 404.	`account-website.e2e.ts`; `modules/outbox/outbox.test.ts`	FR-DEV-01, NFR-SEC-06
TC-86	Try the demo	1. On the stall-mode server, signed out, press "Try the demo".	Signed in as the demo gym's owner, on Today, with every owner page in the navigation.	`account-website.e2e.ts`	FR-WEB-04
TC-87	The demo reset	In the API test, a visitor joins the demo gym; then re-seed it as `npm run demo:reset` does, and again through the stall-mode button's route.	Each time, the visitor is gone, and the demo gym is as seeded.	`modules/demo/demo.test.ts`	FR-OPS-03

10.14 Outside the browser

These belong to no flow. Each runs in `npm test`, or as the gate it names.

ID	Test	Steps	Expected result	Covered by	Covers
TC-88	Writes come as JSON from Tenure	Send a write: 1. as text/plain; 2. from another Origin; 3. of 64 KB and one byte.	1: "Send the request as JSON." 2: 403, "This request didn't come from Tenure." 3: 413, "That request is over 64 KB. Send 64 KB or less.", before anything reads it. A matching Origin passes.	`platform/http/guards.test.ts`	NFR-SEC-02, NFR-SEC-05

ID	Test	Steps	Expected result	Covered by	Covers
TC-89	Rate limits	1. Fail to log in to one account 11 times in 10 minutes. 2. Sign up 6 times from one device in 10 minutes. 3. Ask for reset links past the limit. 4. Send the kiosk 31 requests in a minute.	1: the 11th gets "Too many attempts. Try again in 10 minutes." 2: the 6th gets "Too many sign-ups from this device. Try again in 10 minutes." 3: "Too many reset requests. Try again in 10 minutes." 4: the 31st gets "Too many tries. Try again in a minute."	`modules/auth/auth.test.ts`; `modules/reset/reset.test.ts`; `modules/kiosk/kiosk.test.ts`; `platform/rate-limit.test.ts`	NFR-SEC-03
TC-90	Every route's declaration	Call every route signed out, as a coach and as an owner, and with ids from another gym.	Each route answers as it's declared: a public one to anyone, a signed-in one with 401 "Log in to continue." when signed out, and an owner-only one with 403 for a coach. An id from another gym is not found.	`permissions.test.ts`	NFR-SEC-04, FR-STAFF-06, FR-PRIV-02
TC-91	The stand-ins	Send an email, a text and a payment through the stand-ins, with the network blocked.	Each is kept in its outbox, and none calls out. The email stand-in writes to the outbox table.	`adapters/adapters.test.ts`	FR-DEV-02, NFR-PRIV-01
TC-92	Migrations	Start on a database at each earlier migration, from 0000.	Each upgrades to the latest on start, keeping its rows. A 0000 gym gets the US, and its members no archived date.	`packages/db/src/migrations.test.ts`	FR-OPS-04, NFR-REL-02
TC-93	Backup and restore	1. Run `npm run backup` while the server writes. 2. Stop the server, and run `npm run restore -- <file>`. 3. Restore while the server runs, then restore a file that fails its integrity check.	1: a dated file in `backups/`. 2: the same row counts as at the backup. 3: both refused, with the database left as it was.	The backup test	FR-OPS-02, NFR-REL-03
TC-94	Stall mode	Start the server in stall mode, on 127.0.0.1 and on 0.0.0.0.	It serves the built web app, with a fallback for app routes, and the API's JSON. On 0.0.0.0 the start-up lines give the LAN links for phones and the kiosk. Every e2e flow runs against a stall-mode server.	`stall.test.ts`	FR-OPS-01
TC-95	The demo gym's country	Seed in a zone of each of the five markets, in a zone elsewhere, and with `TENURE_COUNTRY` set.	Each market's zone gives that country, and a zone elsewhere the US. `TENURE_COUNTRY` wins. The sample numbers stay +1.	`seed.test.ts`	FR-GLOBAL-05
TC-96	The five markets	Run the domain tests for each country: the phone rule's table (§7.2.4), the locale and its clock, and the zones.	Every case in §7.2.4 holds. Each market gets English for its country where Intl has it, and en-001 otherwise, each with its own 12- or 24-hour clock. On Node all five have their own English, so the fallback is checked with a country that has none, such as Saudi Arabia.	`packages/domain/src/phone.test.ts`; `packages/domain/src/locale.test.ts`; `packages/domain/src/zones.test.ts`	NFR-LOC-01, FR-GLOBAL-01, FR-GLOBAL-02, FR-GLOBAL-03

ID	Test	Steps	Expected result	Covered by	Covers
TC-97	Contrast	Check every token pair in both themes.	Every colour pair meets WCAG 2.1 AA, and every line and focus ring 3:1.	`packages/ui/src/contrast.test.ts`	NFR-USE-01
TC-98	The gates, on both engines	Run `npm run typecheck`, `npm run lint`, `npm test`, `npm run build` and `npm run e2e` at each phase's merge, and the e2e twice in a row at the end.	All five pass. The lint includes the boundary rules. Every flow passes on every Chrome and WebKit project.	The gate runner	NFR-MAINT-01, NFR-MAINT-02, NFR-COMP-01
TC-99	The trace	Run the trace test over this spec and every test's name.	It passes only when every Must in §3 has a test whose name starts with its ID, or a check in §10.15.	The trace test	NFR-MAINT-03

10.15 Checked by a person

These Musts, or parts of them, need a person. The orchestrator makes the checks at each pause, unless a line says otherwise.

Requirement	What a test can't show	How it's checked
FR-KIOSK-07	Whether the blend looks as picked	The kiosk in both themes, in Chrome and in iPhone Safari on the Simulator, against the patterns in `docs/milestone-2/kiosk-blend.md`
FR-NUDGE-02, FR-NUDGE-03	The hand-off into another app, which Playwright can't follow	In iPhone Safari on the Simulator, "Text message" opens Messages with the number and the message, and the nudge is recorded. The Simulator has no WhatsApp, so WhatsApp needs a real phone, which the plan doesn't schedule yet (§9)
FR-OPS-01, FR-GYM-03	The laptop's network, and Docker	Run `scripts/stall.sh`. In iPhone Safari on the Simulator, the phone and kiosk links from Settings, at the laptop's LAN address, open Tenure and the kiosk
FR-OPS-02	The Docker volume	`scripts/backup.sh` backs up and restores the `tenure-data` volume, with the same data after the round trip
FR-AUTH-08	The command in Docker	In the container, `npm run reset-link -- <email>` prints a link that opens "Choose a new password"
NFR-COMP-01	Real Safari	The core flows clicked through in iPhone Safari on the Simulator, against the stall server, and in Chrome, in both themes at desktop and phone width
NFR-USE-05	Every place colour is used	At each pause, every coloured mark also has a dot, dash, marker, word or edge
NFR-REL-04	The container's health	After `scripts/stall.sh`, `docker compose ps` shows the container healthy
NFR-MAINT-04	The lockfile	The only package added since milestone 1 is `@playwright/test`, as a test-only dependency of `apps/e2e`
NFR-PRIV-02	The runbook	`docs/DEMO.md` says that older backups still hold deleted members
NFR-SEC-06, NFR-PRIV-01	The config	The built web app and its environment hold no secret, and nothing in the code or its dependencies calls an outside service

Appendix A. Traceability matrix

Built from §3, §4, §5 and §10. A use case's requirements are its Requirements row, and a requirement's tests are the tests whose Covers column names it, so a use case's tests are every test that covers one of its requirements. "Checked by a person" points to §10.15.

§3 holds 110 requirements: 104 Musts, 4 Shoulds and 2 Won'ts. §4 holds 28. Every Must has at least one test, and every requirement in §4 has a test or a check by a person. The Shoulds have no test yet: each gets one if it's built.

A.1 Use cases to requirements and tests

Use case	Requirements	Tests
UC-01 Sign up and import members	FR-AUTH-01 to FR-AUTH-03, FR-AUTH-05, FR-AUTH-09, FR-TODAY-01, FR-TODAY-03, FR-BOARD-01, FR-TPL-03, FR-IMPORT-01 to FR-IMPORT-10, FR-WEB-01 to FR-WEB-03, FR-GLOBAL-02, FR-LIVE-03, FR-OPS-01, NFR-SEC-03	TC-02, TC-05, TC-15, TC-18, TC-20, TC-29, TC-53, TC-63 to TC-72, TC-79 to TC-81, TC-83, TC-89, TC-94, TC-96; checked by a person: FR-OPS-01
UC-02 Check in or join at the kiosk	FR-GYM-03, FR-MEMBER-06, FR-TODAY-04, FR-BOARD-03, FR-KIOSK-01 to FR-KIOSK-06, FR-KIOSK-08, FR-GLOBAL-02, FR-LIVE-01, FR-LIVE-02, NFR-PERF-02, NFR-PERF-03, NFR-SEC-03	TC-04 to TC-09, TC-11 to TC-14, TC-16, TC-29, TC-58, TC-75, TC-76, TC-89, TC-96; checked by a person: FR-GYM-03
UC-03 See who's slipping and contact them	FR-STAFF-07, FR-MEMBER-04, FR-TODAY-01, FR-TODAY-02, FR-BOARD-02, FR-NUDGE-01 to FR-NUDGE-08, FR-TPL-01, FR-TPL-02, FR-LIVE-01	TC-02, TC-12, TC-14, TC-16, TC-18 to TC-20, TC-22, TC-31 to TC-38, TC-44, TC-45, TC-76; checked by a person: FR-NUDGE-02, FR-NUDGE-03
UC-04 Work the board and Members	FR-STAFF-06, FR-STAFF-07, FR-MEM-01 to FR-MEM-06, FR-MEMBER-01 to FR-MEMBER-05, FR-TODAY-05, FR-BOARD-01 to FR-BOARD-03, FR-SHELL-01, FR-SHELL-02, FR-SHELL-05, FR-GLOBAL-04, FR-LIVE-01, FR-LIVE-02	TC-06, TC-07, TC-12 to TC-14, TC-16, TC-20 to TC-22, TC-24 to TC-32, TC-37, TC-38, TC-40, TC-45 to TC-47, TC-49, TC-50, TC-65, TC-72, TC-74, TC-76, TC-90
UC-05 Manage staff	FR-AUTH-06, FR-STAFF-01 to FR-STAFF-07, FR-SHELL-04, FR-LIVE-01, FR-LIVE-04, FR-DEV-01, FR-DEV-02	TC-12, TC-14, TC-16, TC-17, TC-22, TC-31, TC-37, TC-38, TC-40, TC-45, TC-49, TC-59 to TC-62, TC-72, TC-76, TC-85, TC-90, TC-91
UC-06 Read Reports	FR-STAFF-06, FR-MEMBER-07, FR-NUDGE-07, FR-REPORT-01 to FR-REPORT-03, FR-BILL-01, FR-SHELL-03 to FR-SHELL-05	TC-22, TC-31, TC-38, TC-40 to TC-43, TC-45, TC-48 to TC-50, TC-72, TC-77, TC-78, TC-90
UC-07 Archive, restore and delete a member	FR-STAFF-07, FR-MEM-01, FR-MEMBER-05 to FR-MEMBER-07, FR-KIOSK-02, FR-KIOSK-06, FR-IMPORT-09, FR-REPORT-01, FR-LIVE-01, FR-PRIV-03, NFR-PRIV-02	TC-05, TC-07, TC-08, TC-12, TC-14, TC-16, TC-22, TC-24, TC-31, TC-37, TC-38, TC-40, TC-42, TC-71, TC-74 to TC-78; checked by a person: NFR-PRIV-02
UC-08 Edit Settings	FR-GYM-01 to FR-GYM-05, FR-STAFF-06, FR-GLOBAL-01 to FR-GLOBAL-03, FR-OPS-01	TC-05, TC-18, TC-22, TC-29, TC-31, TC-40, TC-45, TC-49, TC-56 to TC-58, TC-72, TC-80, TC-90, TC-94, TC-96; checked by a person: FR-GYM-03, FR-OPS-01
UC-09 Reset a password	FR-AUTH-04, FR-AUTH-07, FR-AUTH-08, FR-DEV-01, FR-DEV-02, NFR-SEC-03, NFR-SEC-06	TC-01, TC-60, TC-82, TC-84, TC-85, TC-89, TC-91; checked by a person: FR-AUTH-08, NFR-SEC-06

Use case	Requirements	Tests
UC-10 Run the stall demo	FR-TODAY-01, FR-KIOSK-07, FR-WEB-04, FR-GLOBAL-05, FR-OPS-01 to FR-OPS-03, FR-OPS-05, NFR-PRIV-02	TC-02, TC-04, TC-10, TC-18, TC-78, TC-86, TC-87, TC-93 to TC-95; checked by a person: FR-KIOSK-07, FR-OPS-01, FR-OPS-02, NFR-PRIV-02

A.2 Functional requirements to use cases and tests

Requirement	Priority	Use cases	Tests
FR-AUTH-01	Must	UC-01	TC-80, TC-81
FR-AUTH-02	Must	UC-01	TC-80, TC-81
FR-AUTH-03	Must	UC-01	TC-80
FR-AUTH-04	Must	UC-09	TC-01, TC-82
FR-AUTH-05	Must	UC-01	TC-53, TC-83
FR-AUTH-06	Must	UC-05	TC-17, TC-62
FR-AUTH-07	Must	UC-09	TC-84
FR-AUTH-08	Must	UC-09	TC-84; checked by a person (§10.15)
FR-AUTH-09	Should	UC-01	None yet
FR-AUTH-10	Won't	None	Not in this version
FR-GYM-01	Must	UC-08	TC-56, TC-57
FR-GYM-02	Must	UC-08	TC-57
FR-GYM-03	Must	UC-02, UC-08	TC-58; checked by a person (§10.15)
FR-GYM-04	Must	UC-08	TC-58
FR-GYM-05	Should	UC-08	None yet
FR-STAFF-01	Must	UC-05	TC-60, TC-61
FR-STAFF-02	Must	UC-05	TC-59, TC-62
FR-STAFF-03	Must	UC-05	TC-60, TC-61
FR-STAFF-04	Must	UC-05	TC-60, TC-61
FR-STAFF-05	Must	UC-05	TC-62
FR-STAFF-06	Must	UC-04, UC-05, UC-06, UC-08	TC-22, TC-31, TC-40, TC-45, TC-49, TC-72, TC-90
FR-STAFF-07	Must	UC-03, UC-04, UC-05, UC-07	TC-14, TC-22, TC-31, TC-37, TC-38, TC-76
FR-MEM-01	Must	UC-04, UC-07	TC-24, TC-74
FR-MEM-02	Must	UC-04	TC-21
FR-MEM-03	Must	UC-04	TC-20, TC-25
FR-MEM-04	Must	UC-04	TC-25
FR-MEM-05	Must	UC-04	TC-26
FR-MEM-06	Must	UC-04	TC-27
FR-MEMBER-01	Must	UC-04	TC-28

Requirement	Priority	Use cases	Tests
FR-MEMBER-02	Must	UC-04	TC-30
FR-MEMBER-03	Must	UC-04	TC-29
FR-MEMBER-04	Must	UC-03, UC-04	TC-32
FR-MEMBER-05	Must	UC-04, UC-07	TC-74, TC-76
FR-MEMBER-06	Must	UC-02, UC-07	TC-75, TC-76
FR-MEMBER-07	Must	UC-06, UC-07	TC-42, TC-77, TC-78
FR-TODAY-01	Must	UC-01, UC-03, UC-10	TC-02, TC-18
FR-TODAY-02	Must	UC-03	TC-19, TC-32
FR-TODAY-03	Must	UC-01	TC-63
FR-TODAY-04	Must	UC-02	TC-12
FR-TODAY-05	Should	UC-04	None yet
FR-BOARD-01	Must	UC-01, UC-04	TC-20
FR-BOARD-02	Must	UC-03, UC-04	TC-20, TC-32
FR-BOARD-03	Must	UC-02, UC-04	TC-13
FR-NUDGE-01	Must	UC-03	TC-32
FR-NUDGE-02	Must	UC-03	TC-33, TC-34; checked by a person (§10.15)
FR-NUDGE-03	Must	UC-03	TC-33, TC-34; checked by a person (§10.15)
FR-NUDGE-04	Must	UC-03	TC-35
FR-NUDGE-05	Must	UC-03	TC-36
FR-NUDGE-06	Must	UC-03	TC-38
FR-NUDGE-07	Must	UC-03, UC-06	TC-38
FR-NUDGE-08	Must	UC-03	TC-16, TC-37
FR-TPL-01	Must	UC-03	TC-44, TC-45
FR-TPL-02	Must	UC-03	TC-32, TC-44
FR-TPL-03	Must	UC-01	TC-80
FR-KIOSK-01	Must	UC-02	TC-04
FR-KIOSK-02	Must	UC-02, UC-07	TC-05, TC-07, TC-08
FR-KIOSK-03	Must	UC-02	TC-06
FR-KIOSK-04	Must	UC-02	TC-06
FR-KIOSK-05	Must	UC-02	TC-05, TC-06, TC-11
FR-KIOSK-06	Must	UC-02, UC-07	TC-08
FR-KIOSK-07	Must	UC-10	TC-04, TC-10; checked by a person (§10.15)
FR-KIOSK-08	Must	UC-02	TC-09
FR-KIOSK-09	Won't	None	Not in this version
FR-IMPORT-01	Must	UC-01	TC-64, TC-72

Requirement	Priority	Use cases	Tests
FR-IMPORT-02	Must	UC-01	TC-64
FR-IMPORT-03	Must	UC-01	TC-65
FR-IMPORT-04	Must	UC-01	TC-66
FR-IMPORT-05	Must	UC-01	TC-67
FR-IMPORT-06	Must	UC-01	TC-68
FR-IMPORT-07	Must	UC-01	TC-70
FR-IMPORT-08	Must	UC-01	TC-64, TC-69
FR-IMPORT-09	Must	UC-01, UC-07	TC-71
FR-IMPORT-10	Must	UC-01	TC-15, TC-69
FR-REPORT-01	Must	UC-06, UC-07	TC-40, TC-42
FR-REPORT-02	Must	UC-06	TC-41
FR-REPORT-03	Must	UC-06	TC-40, TC-42
FR-BILL-01	Must	UC-06	TC-43
FR-ME-01	Must	None	TC-53
FR-SHELL-01	Must	UC-04	TC-46
FR-SHELL-02	Must	UC-04	TC-47
FR-SHELL-03	Must	UC-06	TC-48
FR-SHELL-04	Must	UC-05, UC-06	TC-49
FR-SHELL-05	Must	UC-04, UC-06	TC-50
FR-SHELL-06	Must	None	TC-51
FR-SHELL-07	Must	None	TC-52
FR-WEB-01	Must	UC-01	TC-79
FR-WEB-02	Must	UC-01	TC-79
FR-WEB-03	Must	UC-01	TC-79
FR-WEB-04	Must	UC-10	TC-86
FR-WEB-05	Must	None	TC-79
FR-GLOBAL-01	Must	UC-08	TC-57, TC-80, TC-96
FR-GLOBAL-02	Must	UC-01, UC-02, UC-08	TC-05, TC-29, TC-96
FR-GLOBAL-03	Must	UC-08	TC-18, TC-56, TC-96
FR-GLOBAL-04	Must	UC-04	TC-06, TC-07, TC-25, TC-65
FR-GLOBAL-05	Must	UC-10	TC-95
FR-GLOBAL-06	Must	None	TC-55
FR-LIVE-01	Must	UC-02, UC-03, UC-04, UC-05, UC-07	TC-12, TC-16
FR-LIVE-02	Must	UC-02, UC-04	TC-14
FR-LIVE-03	Must	UC-01	TC-15, TC-69

Requirement	Priority	Use cases	Tests
FR-LIVE-04	Must	UC-05	TC-17
FR-PRIV-01	Must	None	TC-54
FR-PRIV-02	Must	None	TC-37, TC-90
FR-PRIV-03	Must	UC-07	TC-77
FR-OPS-01	Must	UC-01, UC-08, UC-10	TC-94; checked by a person (§10.15)
FR-OPS-02	Must	UC-10	TC-93; checked by a person (§10.15)
FR-OPS-03	Must	UC-10	TC-87
FR-OPS-04	Must	None	TC-92
FR-OPS-05	Should	UC-10	None yet
FR-DEV-01	Must	UC-05, UC-09	TC-85
FR-DEV-02	Must	UC-05, UC-09	TC-60, TC-91

A.3 Non-functional requirements to use cases and tests

Requirement	Use cases	Tests
NFR-PERF-01	None	TC-03, TC-20, TC-24, TC-28, TC-38, TC-40, TC-43, TC-44, TC-56, TC-79
NFR-PERF-02	UC-02	TC-05
NFR-PERF-03	UC-02	TC-12
NFR-PERF-04	None	TC-73
NFR-USE-01	None	TC-97
NFR-USE-02	None	TC-09, TC-23, TC-39, TC-51
NFR-USE-03	None	TC-10, TC-39, TC-46, TC-47
NFR-USE-04	None	TC-11, TC-13, TC-52
NFR-USE-05	None	TC-21, TC-41; checked by a person (§10.15)
NFR-SEC-01	None	TC-83
NFR-SEC-02	None	TC-88
NFR-SEC-03	UC-01, UC-02, UC-09	TC-89
NFR-SEC-04	None	TC-90
NFR-SEC-05	None	TC-72, TC-88
NFR-SEC-06	UC-09	TC-85; checked by a person (§10.15)
NFR-PRIV-01	None	TC-36, TC-54, TC-91; checked by a person (§10.15)
NFR-PRIV-02	UC-07, UC-10	TC-78; checked by a person (§10.15)
NFR-REL-01	None	TC-17
NFR-REL-02	None	TC-92
NFR-REL-03	None	TC-93
NFR-REL-04	None	checked by a person (§10.15)

Requirement	Use cases	Tests
NFR-COMP-01	None	TC-98; checked by a person (§10.15)
NFR-MAINT-01	None	TC-98
NFR-MAINT-02	None	TC-98
NFR-MAINT-03	None	TC-99
NFR-MAINT-04	None	checked by a person (§10.15)
NFR-LOC-01	None	TC-96
NFR-COPY-01	None	TC-55

Appendix B. Open issues

Each issue says what's open and what waits on it. Uzair settles the product questions, and his answers go into `docs/milestone-2/answers.md`. The orchestrator settles the spec's own fixes, and builders the technical calls. An issue leaves this list once it's settled and the spec says so.

B.1 Waiting on Uzair

ID	Issue	What's open	What waits on it
OI-01	The design polish picks	The six design reviewers run again, a synthesizer writes `design-review.md`, and the picks page "Tenure design polish" is drawn and checked. Uzair picks the taste calls by eye, and the harmony errors go straight in, listed (§8.7)	UI2 and every phase 2 screen job. Phase 2 starts after the picks (R-02)
OI-02	The core-flow edge cases	All eight finders have reported, 112 cases in all. The synthesizer's `docs/milestone-2/edge-cases.md`, written on 7 Nov and not yet committed, sorts them into 13 questions for Uzair, technical calls for builders, bug risks with the tests that pin them, and a later-polish list. Uzair answers the questions in rounds of four, the recommended option first (§8.7)	Phase 1's branches, which stay unmerged until the answers fold in (R-05)
OI-03	The home page words	A copywriter offers 2 or 3 options for the headline, the line under it and the buttons, in the voice of `marketing/brand/DESIGN.md`, and Uzair picks (§8.7)	C4 and WEB in phase 4 (FR-WEB-01, TC-79)
OI-04	The product's name	"Tenure" is a working name. The final name needs a trademark and web domain check in the five markets. In the interface it's one copy id, `brand.name`, and a few other lines of COPY.md, so the words change in one place. Code names such as the `tenure_session` cookie, `TENURE_COUNTRY` and the `tenure-data` volume needn't change	The website going online, with hosting in the hardening. Better settled before the home page copy round (OI-03)
OI-05	Legal steps per market	Which steps a gym needs before it enters its real members, and which can wait for hosting, real providers and payments in the hardening. B.2 gives the starting points	Each market's first real gym
OI-06	The price	No price shows anywhere until Uzair sets one (FR-BILL-01, FR-WEB-02). Open: the price and its plans, the currency in each market, and when Billing's plan button comes. The old app showed $99 a month with unlimited members (`legacy/app/SPEC.md`). Payments need a provider for every market, and Stripe doesn't take Pakistan-based businesses (PROGRESS.md)	A price on Pricing and Billing, and payments in the hardening
OI-07	Copying at the stall	Phones reach the app at the laptop's LAN address over plain HTTP. That isn't a secure context, so browsers withhold `navigator.clipboard`, and "Copy message" and the Copy buttons for the kiosk, phone and invite links fail there. The old app had a fallback in Settings only. Nothing in §3 or §4 asks for copying to work at the stall, and Playwright on 127.0.0.1 can't catch it. Proposed: each Copy falls back to selecting the text for the user to copy, FR-NUDGE-02 says so, and §10.15 checks it on the Simulator against the stall server	W-CONTACT, W-SETTINGS and W-STAFF (FR-NUDGE-02, FR-GYM-03, FR-STAFF-03)
OI-08	The demo reset button	Where the stall-mode button sits, who may press it, what it says (COPY.md has no words for it yet), and what happens to signed-in demo sessions. As it stands, any visitor signed in as the demo owner (FR-WEB-04) could reset the demo in the middle of someone else's. `edge-cases.md` Q10 asks where it sits and who may press it; its words stay C0's question	WEB in phase 4 (FR-OPS-03, TC-87)

ID	Issue	What's open	What waits on it
OI-09	The reset request limit	NFR-SEC-03 gives no number for reset requests, so TC-89 can check only the answer past the limit. The copy says "Try again in 10 minutes." Proposed: 5 requests per email per 10 minutes	A1 in phase 4 (NFR-SEC-03, TC-89)
OI-10	Import check-ins for an archived member	FR-IMPORT-09 restores an archived member on any matching row, but `import.problem.restored` words member rows only. Open: whether a check-in row restores the member, stores the check-in alone, or is skipped (§7.2.6 covers member rows only)	API-IMPORT and W-IMPORT (FR-IMPORT-09, TC-71)
OI-11	The share on the line	FR-TODAY-01 doesn't define it. The domain divides the members on the line by those with at least one finished week, as the old app did (§7.2.1). A member with exactly one finished week is always new, so they can only lower the share. Kept on purpose; Uzair to confirm, then FR-TODAY-01 states it	FR-TODAY-01's wording, and TC-18
OI-12	The invite button's words	"Create link", or "Send invite" now that the invite is emailed. C0's question, asked beside `edge-cases.md` Q12	C3 and W-STAFF in phase 3 (FR-STAFF-03)
OI-13	Keeping spent links and emails	No limit is set on keeping spent invites, used or expired reset links, or outbox emails (§6.3). The outbox holds working invite and reset links as plain text, in the live database and in every backup, and nothing yet requires reset tokens to be stored hashed. Proposed: reset tokens stored as sha256, like invite tokens (NFR-SEC-01), and spent links and outbox emails deleted after 30 days	A0, A1 and A2 (FR-AUTH-07, FR-STAFF-03, FR-DEV-02)

B.2 Legal steps per market

The first version keeps each gym's data on the machine that runs it, sends no message itself and takes no payment. Each line below is a starting point, to confirm with a lawyer in that market (OI-05).

Market	Starting points
All five	Terms for gyms, and a privacy notice. A processing agreement, since each gym controls its members' data and Tenure processes it for them. Where the data is hosted, once hosting comes
US	State privacy laws, which apply by a business's size. Consent and 10DLC registration for automated texts (PROGRESS.md)
Canada	PIPEDA, and Quebec's Law 25 for gyms in Quebec. CASL for commercial texts and email
UK	UK GDPR and the Data Protection Act 2018: the ICO's data protection fee, a lawful basis for members' data, and data-protection requests, which permanent delete serves (FR-MEMBER-07). PECR for texts and email
Pakistan	Whether the national data protection bill has become law. PTA rules for texts (PROGRESS.md). Payments need a company in a Stripe country, or a merchant of record such as Paddle or Lemon Squeezy (PROGRESS.md)
UAE	The federal data protection law (Federal Decree-Law No. 45 of 2021), and the DIFC and ADGM free zones' own laws for gyms there. The local rules for texts

B.3 Requirement fixes waiting on the orchestrator

These are gaps and conflicts in §3 and §4 that this draft leaves as they are, since a requirement's meaning changes only there. Each has a proposed fix, and the use cases and tests already follow it where they name the behaviour. OI-09 and OI-11 above are requirement fixes too.

ID	Requirement	Problem	Proposed fix
OI-14	FR-STAFF-06	Its owner-only list leaves out permanent delete, which FR-MEMBER-07 gives to owners, so the permissions sweep (NFR-SEC-04) has no declaration for it	Add permanent delete to the list. TC-78 checks that a coach's delete is refused
OI-15	FR-STAFF-04, §6.1	The invitee sets only a name and a password, but the new account needs an email. The invite page's copy asks for one (`invite.email`, a field), and the 0000 `invites` table keeps none	Keep FR-STAFF-04, as UC-05 and TC-60 do. The invite keeps the address it was sent to, in a column in migration 0003, and the account takes it; `invite.email` becomes plain text (`edge-cases.md` SA3). The other way, FR-STAFF-04 says "name, email and password"
OI-16	FR-AUTH-01, FR-STAFF-04	Neither says that an email already in use is refused. `users.email` is unique across every gym, ignoring case (§6.1), and `api.duplicate` words the refusal. UC-01 and UC-05 cite §6.1 for it, and `edge-cases.md` Q3 asks what happens when an owner invites an existing coach as a co-owner	Add "An email already in use is refused." to both, and a step for it to TC-81 and TC-60
OI-17	FR-AUTH-07	Gives no length rule for the new password, and doesn't say whether saving it signs the user in. The field's help says "8 characters or more.", and UC-09 5b relies on it	"The new password is 8 to 200 characters, as at sign-up", and where the user lands after "Password saved"
OI-18	FR-MEMBER-05	Nothing says an archived member leaves Today, the board, and Members' "First 90 days" and "Everyone", though ARCHITECTURE §2, answers.md round 3b and §1.3 do. TC-74 tests it, and §7.2.3 cites §1.3 for it	Add: an archived member leaves every list except "Archived", and the kiosk's name search, and their history stays
OI-19	FR-MEMBER-07, FR-REPORT-01	Neither says that Reports stop counting a deleted member, as ARCHITECTURE §2 does. UC-06, §6.3 and TC-42 rely on it	Add to FR-REPORT-01: a deleted member's history no longer counts
OI-20	FR-LIVE-02	Doesn't say who receives the gym-wide events: an import, a staff change and a settings change. Since commit `41b8077` they reach everyone signed in to the gym, coaches included, and TC-16 tests that	Add: gym-wide events reach everyone signed in to the gym
OI-21	FR-GYM-05, FR-LIVE-01	A settings change's event is a Should in FR-GYM-05 and a Must in FR-LIVE-01. FIRST-VERSION.md's approved calls include it	Drop FR-GYM-05, or make it a Must. FR-LIVE-01 and TC-16 cover it
OI-22	FR-OPS-05	A Should, but NFR-PRIV-02 is verified partly by DEMO.md, which OPS2 writes in phase 4	Make FR-OPS-05 a Must
OI-23	FR-KIOSK-08	Covers only the 30 s idle return. The welcome's own countdown back to the start ("Ready for the next member in {n}.", from 5 to 1) has no requirement, and UC-02 step 5 uses it	Add: after a welcome, the kiosk counts down from 5 and returns to its start
OI-24	FR-TODAY-02	No requirement gives Today's rows their "Open" button, though ARCHITECTURE §2 "Links", answers.md F3 and W-TODAY-BOARD do	Add: each row ends with "Open", as in Members (FR-MEM-03)

ID	Requirement	Problem	Proposed fix
OI-25	§3.7, FR-REPORT-01	No Must states the week, pace and status rules, or the cohort and visits-by-week maths: they're only in §1.3, §7.2 and ARCHITECTURE §7. Their domain tests carry no ID, so the trace test (NFR-MAINT-03) has no Must for them to name	A new Must in §3.7: weeks, pace and status follow §7.2.1. FR-REPORT-01 cites §7.2.7
OI-26	FR-MEM-04	"Members with no last name come first" can be read as every single name going to the top. The rule in `docs/milestone-2/jobs/M2.md` is the first name, then the last name with no last name first, then the id	Say so, with its example: "Sam" comes before "Sam O.". TC-25 tests that rule
OI-27	FR-MEM-05	Doesn't say whether the phone is optional in "Add a member"	Optional, as at the kiosk join. FR-NUDGE-04 covers a member with no phone
OI-28	NFR-PERF-03	Verified "with two browsers", but each engine has its own server and database, so two engines never share data	"with two browser contexts in one engine", as TC-12 has it

B.4 Outside this spec

ID	Where	What's open	Who
OI-29	`packages/copy/COPY.md`	`status.below` says "1 or 2 visits in the last finished week, and not slipping", but finished weeks of 2 then 0 are below the line with no visits (§7.2.1, ARCHITECTURE §7). Under NFR-COPY-01 the wrong legend line would ship	C2, in phase 2
OI-30	`docs/ARCHITECTURE.md`	§1 still lists email verification among the account basics, against §2, §13 and FR-AUTH-10. §2's "Home-screen kiosk" row and §3's line on `legacy/app` still say milestone 5, but §13 now ends with the hardening, milestone 4	A contract commit
OI-31	`apps/e2e/playwright.config.ts`	It runs `kiosk*.e2e.ts` on the iPad only, so the kiosk flow never runs on Chrome. §8.3, §10 and FIRST-VERSION.md's phase 2 have it pass on the phone projects too	Phase 2's owner of the e2e fixtures
OI-32	§9	The mitigations of R-12, R-13 and R-15 are proposed, not yet in the plan. R-15's is the only check of WhatsApp on a real phone	Uzair, with the orchestrator
OI-33	§6.1	The `members.source` value for a member added by hand isn't named yet	API-MEMBERS
OI-34	`docs/milestone-2/answers.md`	Round 5 says en-001 covers the UAE, but Node resolves en-AE (`edge-cases.md` GL7), so the UAE gets its own English. FR-GLOBAL-03's rule holds either way, and TC-96 no longer claims the UAE falls back	The orchestrator